Integrable Systems

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Most differential equations cannot be solved. This is not a statement about our own cleverness – it is a statement about differential equations, the overwhelming majority of which have solutions that admit no description more economical than the equation itself. Every so often, though, we stumble across an equation that can be 'integrated up': one whose solutions can be written down in closed form, or reduced to quadratures, or generated from nothing by an algebraic recipe. Such equations are called *integrable*, and they are the subject of this course.

There is a curious tension here. On the one hand, integrable systems ought to be vanishingly rare, in the same way that a randomly chosen polynomial is unlikely to factorise nicely. On the other hand, integrable equations turn up all over physics – the harmonic oscillator, the Kepler problem, the spinning top, waves in shallow water, light in an optical fibre. Nature seems unreasonably fond of them. The explanation, when we find it, is always the same: integrability is a symptom of hidden structure. An integrable system has more conserved quantities than it has any right to have, and those conserved quantities conspire to fill up phase space with tori on which the motion is nothing more than uniform rotation.

The plan of the course is to take this idea and push it as far as it will go. We begin with ordinary differential equations, where the story is essentially complete: the Arnold–Liouville theorem tells us precisely what it means for a Hamiltonian system to be integrable, and how to integrate it. We then move to partial differential equations, which we can think of as infinite-dimensional dynamical systems. Here the theory is much wilder and much more beautiful. We will meet the Korteweg–de Vries equation and its *solitons* – solitary waves that collide and emerge unscathed, behaving far more like particles than like waves – and we will spend the heart of the course developing the *inverse scattering transform*, an extraordinary machine which solves a nonlinear PDE by turning it into a linear scattering problem, letting the scattering data evolve trivially, and then reconstructing the answer.

This article constitutes my notes for the 'Integrable Systems' course, held in Michaelmas 2021 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered[1].

Contents

[1] We will lean fairly heavily on IB Variational Principles (Hamilton's equations, the Legendre transform, Noether's theorem) and on IB Methods (Fourier transforms, Green's functions, Sturm–Liouville theory). A passing acquaintance with IB Quantum Mechanics – specifically reflection and transmission coefficients for a one-dimensional potential – will make Section 3 feel much more familiar.

§1 Integrability of Ordinary Differential Equations

We begin with ordinary differential equations. The eventual goal of this section is the Arnold–Liouville theorem, which says that a Hamiltonian system possessing enough conserved quantities can be solved completely, and moreover tells us how. Before we get there we need to set up some machinery: flows, Poisson brackets, and canonical transformations.

§1.1 Vector Fields and Flow Maps

An m-dimensional first-order ODE is specified by a **vector field** $\mathbf{V} : \mathbb{R}^m \to \mathbb{R}^m$ together with an initial condition $\mathbf{x}_0 \in \mathbb{R}^m$. We look for a curve $\mathbf{x}(t) \in \mathbb{R}^m$, defined for t in some interval containing 0, satisfying

$$\dot{\mathbf{x}} = \mathbf{V}(\mathbf{x}), \qquad \mathbf{x}(0) = \mathbf{x}_0.$$

Restricting to first-order equations costs us nothing: an nth order ODE in one variable becomes a first-order system in n variables in the usual way, by treating the derivatives as new coordinates.

Throughout this course we will be cavalier about regularity, and assume that all our vector fields are as nice as we need them to be.

Fact 1.1

For a sufficiently nice vector field $\mathbf{V}$ and any initial condition $\mathbf{x}_0$, the problem $\dot{\mathbf{x}} = \mathbf{V}(\mathbf{x})$, $\mathbf{x}(0) = \mathbf{x}_0$ has a unique solution, and that solution depends smoothly on both t and $\mathbf{x}_0$.

Because the solution is unique, it is legitimate to package all of the solutions together into a single object.

Definition 1.2 (Flow map)

The **flow map** of a vector field $\mathbf{V}$ is the family of maps $g^t : \mathbb{R}^m \to \mathbb{R}^m$ defined by

$$\mathbf{x}(t) = g^t\mathbf{x}_0,$$

where $\mathbf{x}(t)$ solves $\dot{\mathbf{x}} = \mathbf{V}(\mathbf{x})$ with $\mathbf{x}(0) = \mathbf{x}_0$.

The flow map is smooth, and satisfies exactly the properties one would hope for.

Proposition 1.3

The flow map obeys

(i) $g^0 = \mathrm{id}$;

(ii) $g^{t+s} = g^t g^s$;

(iii) $(g^t)^{-1} = g^{-t}$.

Proof. The first is immediate from the definition, and the third follows from the first two on setting $s = -t$. For the second, fix $\mathbf{x}_0$ and s, and regard both $t \mapsto g^{t+s}\mathbf{x}_0$

and $t \mapsto g^t(g^s\mathbf{x}_0)$ as functions of t. Both solve

$$\dot{\mathbf{x}} = \mathbf{V}(\mathbf{x}), \qquad \mathbf{x}(0) = g^s\mathbf{x}_0,$$

and so by uniqueness of solutions they coincide. □

In the language of IA Groups, this says that $t \mapsto g^t$ is a homomorphism from $(\mathbb{R}, +)$ into the group of diffeomorphisms of $\mathbb{R}^m$. We say that $\mathbf{V}$ is the **infinitesimal generator** of the flow, which is justified by Taylor expanding:

$$g^\varepsilon\mathbf{x}_0 = \mathbf{x}(0) + \varepsilon\dot{\mathbf{x}}(0) + o(\varepsilon) = \mathbf{x}_0 + \varepsilon\mathbf{V}(\mathbf{x}_0) + o(\varepsilon).$$

So to first order in ε, flowing for time ε is just moving along $\mathbf{V}$.

Suppose now that we have two vector fields $\mathbf{V}_1, \mathbf{V}_2$ generating flows g_1^t and g_2^s. A natural question is whether the flows *commute*, that is, whether

$$g_1^t g_2^s \mathbf{x}_0 = g_2^s g_1^t \mathbf{x}_0$$

for all $\mathbf{x}_0, s, t$. Asked directly this is a hopeless question, since it requires us to solve both ODEs. But there is an infinitesimal criterion, and infinitesimal criteria are always easier.

Definition 1.4 (Commutator)
Given vector fields $\mathbf{V}_1, \mathbf{V}_2 : \mathbb{R}^m \to \mathbb{R}^m$, their **commutator** is the vector field

$$[\mathbf{V}_1, \mathbf{V}_2] = \left(\mathbf{V}_1 \cdot \frac{\partial}{\partial \mathbf{x}}\right)\mathbf{V}_2 - \left(\mathbf{V}_2 \cdot \frac{\partial}{\partial \mathbf{x}}\right)\mathbf{V}_1,$$

where $\frac{\partial}{\partial \mathbf{x}} = (\partial_{x_1}, \dots, \partial_{x_m})^T$. Componentwise,

$$[\mathbf{V}_1, \mathbf{V}_2]_i = \sum_{j=1}^m \left((\mathbf{V}_1)_j \frac{\partial(\mathbf{V}_2)_i}{\partial x_j} - (\mathbf{V}_2)_j \frac{\partial(\mathbf{V}_1)_i}{\partial x_j}\right).$$

Proposition 1.5
Let $\mathbf{V}_1, \mathbf{V}_2$ be vector fields generating flows g_1^t, g_2^s. Then

$$[\mathbf{V}_1, \mathbf{V}_2] = 0 \iff g_1^t g_2^s = g_2^s g_1^t \text{ for all } s, t.$$

We will not prove this (it is a standard exercise, and appears on the first example sheet), but the idea is worth recording: expanding $g_1^{-t} g_2^{-s} g_1^t g_2^s \mathbf{x}_0$ to second order in the small parameters, everything cancels except a term $st\,[\mathbf{V}_1, \mathbf{V}_2](\mathbf{x}_0)$. The commutator measures precisely the failure of the two flows to commute, and it does so at the level of the vector fields, where we can compute.

§1.2 Hamiltonian Systems

We now specialise to a very particular class of ODEs. In IB Variational Principles we met Hamilton's equations, obtained from a Lagrangian by a Legendre transform; here we take them as our starting point.

The setting is a **phase space** $M = \mathbb{R}^{2n}$, with coordinates

$$(\mathbf{q}, \mathbf{p}) = (q_1, \ldots, q_n, p_1, \ldots, p_n),$$

which we think of as generalised positions and generalised momenta. We will usually write

$$\mathbf{x} = (\mathbf{q}, \mathbf{p})^T \in M.$$

The crucial point is that M is not just $\mathbb{R}^{2n}$: the coordinates come *paired up*, with q_i married to p_i. This pairing is the extra structure that distinguishes a phase space from a bare vector space, and we encode it in the $2n \times 2n$ antisymmetric matrix

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix},$$

called the **symplectic form**. Note that $J^T = -J$ and $J^2 = -I_{2n}$.

Almost everything in this section can be phrased in terms of J alone. The first example is the Poisson bracket.

Definition 1.6 (Poisson bracket)

For smooth $f, g : M \to \mathbb{R}$, the **Poisson bracket** is

$$\{f, g\} = \frac{\partial f}{\partial \mathbf{x}} \cdot J \frac{\partial g}{\partial \mathbf{x}} = \frac{\partial f}{\partial \mathbf{q}} \cdot \frac{\partial g}{\partial \mathbf{p}} - \frac{\partial f}{\partial \mathbf{p}} \cdot \frac{\partial g}{\partial \mathbf{q}}.$$

The two expressions agree by direct computation with the block form of J. The Poisson bracket has a list of properties, most of which are obvious and one of which is not.

Proposition 1.7

The Poisson bracket satisfies, for all smooth $f, g, h : M \to \mathbb{R}$:

(i) it is bilinear;

(ii) it is antisymmetric, $\{f, g\} = -\{g, f\}$;

(iii) it obeys the Leibniz rule $\{f, gh\} = \{f, g\}h + \{f, h\}g$;

(iv) it obeys the Jacobi identity

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0;$$

(v) on the coordinate functions,

$$\{q_i, q_j\} = \{p_i, p_j\} = 0, \qquad \{q_i, p_j\} = \delta_{ij}.$$

Proof. All of these are immediate from the definition except the Jacobi identity, which one proves by writing out all twenty-four terms and watching them cancel. The only structural remark worth making is that each of the three double brackets contains second derivatives of exactly one of f, g, h, and these second-derivative terms cancel in pairs because J is antisymmetric and second partial derivatives commute. □

The last of these is called the **canonical commutation relation**, and it is the algebraic shadow of the pairing between q_i and p_i.

Definition 1.8 (Hamilton's equations)

Given a function $H : M \to \mathbb{R}$, called the **Hamiltonian**, **Hamilton's equations** are

$$\dot{\mathbf{q}} = \frac{\partial H}{\partial \mathbf{p}}, \qquad \dot{\mathbf{p}} = -\frac{\partial H}{\partial \mathbf{q}}. \tag{$*$}$$

In terms of J these collapse into the single equation

$$\dot{\mathbf{x}} = J\frac{\partial H}{\partial \mathbf{x}}.$$

This is worth pausing over. To specify a general ODE on $\mathbb{R}^{2n}$ we must supply a vector field, which is $2n$ functions. To specify a Hamiltonian system we need only supply the *single* function H. The symplectic structure does the rest of the work. This is our first hint that Hamiltonian systems are special.

Definition 1.9 (Hamiltonian vector field)

Given $f : M \to \mathbb{R}$, the associated **Hamiltonian vector field** is

$$\mathbf{V}_f = J\frac{\partial f}{\partial \mathbf{x}}.$$

Thus Hamilton's equations say precisely that the motion is the flow of $\mathbf{V}_H$.

Now suppose we are watching a particle move according to $(*)$, and we are interested in some observable $f : M \to \mathbb{R}$ – its energy, its angular momentum, its distance from the origin. How does f change along the motion? The chain rule gives

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}f(\mathbf{x}(t)) = \frac{\partial f}{\partial \mathbf{x}} \cdot \dot{\mathbf{x}} = \frac{\partial f}{\partial \mathbf{x}} \cdot J\frac{\partial H}{\partial \mathbf{x}} = \{f, H\}.$$

We record this, because it is the single most useful formula in the subject.

Proposition 1.10

If $\mathbf{x}(t)$ evolves according to Hamilton's equations with Hamiltonian H, then for any smooth $f : M \to \mathbb{R}$,

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \{f, H\}.$$

In other words – f is conserved if and only if it Poisson commutes with H. This is genuinely powerful. Without it, deciding whether some quantity is conserved means solving the equations of motion and staring at the answer, or else differentiating and hoping that things cancel. With it, we simply compute a bracket. And immediately we get one conserved quantity for free:

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \{H, H\} = 0$$

by antisymmetry. Energy is conserved.

Example 1.11 (Particle in a potential)

Consider a particle of unit mass with Cartesian position $\mathbf{q} = (q_1, q_2, q_3)$ moving in a potential $U(\mathbf{q})$. Newton's second law reads

$$\ddot{\mathbf{q}} = -\frac{\partial U}{\partial \mathbf{q}}.$$

Introducing $\mathbf{p} = \dot{\mathbf{q}}$, we have

$$\dot{\mathbf{x}} = \begin{pmatrix} \dot{\mathbf{q}} \\ \dot{\mathbf{p}} \end{pmatrix} = \begin{pmatrix} \mathbf{p} \\ -\partial U/\partial \mathbf{q} \end{pmatrix} = J\frac{\partial H}{\partial \mathbf{x}}, \qquad H = \tfrac{1}{2}|\mathbf{p}|^2 + U(\mathbf{q}),$$

since $\partial H/\partial \mathbf{p} = \mathbf{p}$ and $\partial H/\partial \mathbf{q} = \partial U/\partial \mathbf{q}$. So Newtonian mechanics in a potential is a Hamiltonian system, and the Hamiltonian is the total energy.

Example 1.12 (Angular momentum in the two-body problem)

Fix the Sun at the origin and let a planet have Cartesian coordinates $\mathbf{q}$. The equation of motion is

$$\ddot{\mathbf{q}} = -\frac{\mathbf{q}}{|\mathbf{q}|^3},$$

which by the previous example is Hamiltonian with $\mathbf{p} = \dot{\mathbf{q}}$ and

$$H = \tfrac{1}{2}|\mathbf{p}|^2 - \frac{1}{|\mathbf{q}|}.$$

The angular momentum is $\mathbf{L} = \mathbf{q} \wedge \mathbf{p}$, or in components $L_i = \varepsilon_{ijk} q_j p_k$. Let us check that it is conserved, using summation convention throughout:

$$\begin{aligned}
\{L_i, H\} &= \frac{\partial L_i}{\partial q_\ell}\frac{\partial H}{\partial p_\ell} - \frac{\partial L_i}{\partial p_\ell}\frac{\partial H}{\partial q_\ell} \\
&= \varepsilon_{ijk}\delta_{j\ell}p_k \cdot p_\ell - \varepsilon_{ijk}q_j\delta_{k\ell} \cdot \frac{q_\ell}{|\mathbf{q}|^3} \\
&= \varepsilon_{ijk}\left(p_j p_k - \frac{q_j q_k}{|\mathbf{q}|^3}\right) \\
&= 0,
\end{aligned}$$

since in each term we are contracting the antisymmetric ε_{ijk} against something symmetric in $j \leftrightarrow k$. So all three components of $\mathbf{L}$ are conserved, as is H: this system has rather a lot of conserved quantities, which is exactly why the Kepler problem can be solved.

There is one more piece of structure worth recording. We now have two ways of combining functions to produce vector fields: we can take the Poisson bracket $\{f, g\}$ and then form its Hamiltonian vector field, or we can form $\mathbf{V}_f$ and $\mathbf{V}_g$ and take their commutator. These agree up to sign.

Proposition 1.13

For smooth $f, g : M \to \mathbb{R}$ we have $[\mathbf{V}_f, \mathbf{V}_g] = -\mathbf{V}_{\{f,g\}}$.

The proof is a computation using the Jacobi identity, and is left to the first example sheet. The content of the statement is that $f \mapsto \mathbf{V}_f$ is (up to a sign) a homomorphism of Lie algebras, from functions-under-Poisson-bracket to vector-fields-under-commutator. We will use it in the proof of the Arnold–Liouville theorem, where it will tell us that flows generated by commuting conserved quantities themselves commute.

§1.3 First Integrals and Liouville Integrability

Definition 1.14 (First integral)
Let (M, H) be a Hamiltonian system. A function $f : M \to \mathbb{R}$ is a **first integral** if

$$\{f, H\} = 0,$$

equivalently, if f is constant along every trajectory.

The terminology is historical. When we solve a differential equation by successive integrations, each integration produces a constant; the constant produced by the first integration was called the first integral. For our purposes, a first integral is simply a constant of the motion.

Our goal for the rest of this section is the following. If a Hamiltonian system has *enough* first integrals, then there is a change of coordinates in which the equations of motion become trivial, and we can integrate them up by inspection. To make 'enough' precise we need two conditions.

Definition 1.15 (Involution)
Two first integrals F, G are **in involution** if $\{F, G\} = 0$. We also say that they **Poisson commute**.

Definition 1.16 (Independence)
Functions $f_1, \dots, f_n : M \to \mathbb{R}$ are **independent** if at each point $\mathbf{x} \in M$ the gradients $\frac{\partial f_i}{\partial \mathbf{x}}$, $i = 1, \dots, n$, are linearly independent vectors in $\mathbb{R}^{2n}$.

Definition 1.17 (Liouville integrable)
A $2n$-dimensional Hamiltonian system (M, H) is **integrable** (in the Liouville sense) if there exist n first integrals $f_1, \dots, f_n$ which are independent and in involution, that is $\{f_i, f_j\} = 0$ for all i, j.

By convention we always take $f_1 = H$, which is legitimate since $\{H, H\} = 0$.

The independence requirement is what stops the definition being vacuous. Without it we could always cheat, taking $f_1 = H$, $f_2 = H^2$, $f_3 = e^H$, and so on: these all Poisson commute with everything in sight, but they carry no information beyond H itself. Independence says that each new first integral genuinely cuts down the available motion.

The involutivity requirement is subtler and less obviously natural, but it is exactly what makes the proof of the Arnold–Liouville theorem work. It is worth noting that a system can have many first integrals without being integrable in this sense, if the integrals refuse to Poisson commute.

Example 1.18

Every two-dimensional Hamiltonian system ($n = 1$) is integrable: we need one first integral, and H itself will do. This is a restatement of the familiar fact that a single particle moving in one dimension in a potential can always be solved by energy conservation and a quadrature.

Example 1.19

Consider the two-body problem above. We have H, and the three components of $\mathbf{L}$. However

$$\{L_1, L_2\} = L_3 \neq 0,$$

so the L_i are not in involution with each other, and we cannot simply take all four. But H, L_3 and $|\mathbf{L}|^2$ *are* pairwise in involution and independent, and the system is six-dimensional, so $n = 3$ and the Kepler problem is integrable.

§1.4 Canonical Transformations

We want to find coordinates in which an integrable system becomes trivial. But we cannot use arbitrary coordinates: the whole structure above depended on the pairing of q_i with p_i, and a general change of variables will destroy it. So we must decide which coordinate changes are allowed. There are several equivalent ways to do this; we take the most pragmatic, and simply demand that Hamilton's equations keep their form.

Definition 1.20 (Canonical transformation)

A coordinate change $(\mathbf{q}, \mathbf{p}) \mapsto (\mathbf{Q}(\mathbf{q}, \mathbf{p}), \mathbf{P}(\mathbf{q}, \mathbf{p}))$ is **canonical** if it preserves the form of Hamilton's equations: that is, if

$$\dot{\mathbf{q}} = \frac{\partial H}{\partial \mathbf{p}}, \quad \dot{\mathbf{p}} = -\frac{\partial H}{\partial \mathbf{q}} \qquad \iff \qquad \dot{\mathbf{Q}} = \frac{\partial \tilde{H}}{\partial \mathbf{P}}, \quad \dot{\mathbf{P}} = -\frac{\partial \tilde{H}}{\partial \mathbf{Q}},$$

where $\tilde{H}(\mathbf{Q}, \mathbf{P}) = H(\mathbf{q}, \mathbf{p})$.

Writing $\mathbf{x} = (\mathbf{q}, \mathbf{p})$ and $\mathbf{y} = (\mathbf{Q}, \mathbf{P})$, this says

$$\dot{\mathbf{x}} = J\frac{\partial H}{\partial \mathbf{x}} \iff \dot{\mathbf{y}} = J\frac{\partial \tilde{H}}{\partial \mathbf{y}}.$$

Example 1.21

Swapping $\mathbf{q}$ and $\mathbf{p}$ is *not* canonical: it introduces a sign error, turning $\dot{\mathbf{Q}} = \partial\tilde{H}/\partial\mathbf{P}$ into $\dot{\mathbf{Q}} = -\partial\tilde{H}/\partial\mathbf{P}$. The map $(\mathbf{q}, \mathbf{p}) \mapsto (\mathbf{p}, -\mathbf{q})$, on the other hand, is canonical.

Let us do the linear case honestly, since it tells us what the general answer must look like.

Example 1.22 (Linear canonical transformations)

Consider $\mathbf{x} \mapsto \mathbf{y} = A\mathbf{x}$ for a constant matrix A. We claim this is canonical if and only if $AJA^T = J$, in which case A is called **symplectic**.

Indeed, by linearity $\dot{\mathbf{y}} = A\dot{\mathbf{x}} = AJ\frac{\partial H}{\partial \mathbf{x}}$. Setting $\tilde{H}(\mathbf{y}) = H(\mathbf{x})$ and using the chain rule,

$$\frac{\partial H}{\partial x_i} = \frac{\partial y_j}{\partial x_i}\frac{\partial \tilde{H}}{\partial y_j} = A_{ji}\frac{\partial \tilde{H}}{\partial y_j} = \left[A^T \frac{\partial \tilde{H}}{\partial \mathbf{y}}\right]_i .$$

Substituting back,

$$\dot{\mathbf{y}} = AJA^T \frac{\partial \tilde{H}}{\partial \mathbf{y}},$$

which is Hamilton's equation for $\tilde{H}$ precisely when $AJA^T = J$.

Since a differentiable map is locally linear, and Hamilton's equations are a local condition, the general statement is the obvious one.

Proposition 1.23

A smooth map $\mathbf{x} \mapsto \mathbf{y}(\mathbf{x})$ is canonical if and only if its Jacobian $D\mathbf{y}$ is symplectic at every point:

$$D\mathbf{y}\, J\, (D\mathbf{y})^T = J.$$

This is a clean criterion, but it is a criterion for *checking* that a transformation is canonical, not for *producing* one. For that we use generating functions.

§1.4.1 Generating Functions

There are four standard flavours of generating function, differing in which pair of old and new variables one takes as independent. We will need only one of them.

Let $S : \mathbb{R}^{2n} \to \mathbb{R}$ be a function which we write as $S(\mathbf{q}, \mathbf{P})$ – old positions and new momenta – and define a transformation implicitly by

$$\mathbf{p} = \frac{\partial S}{\partial \mathbf{q}}, \qquad \mathbf{Q} = \frac{\partial S}{\partial \mathbf{P}}.$$

These equations are to be read as follows. The first is n equations relating $\mathbf{p}, \mathbf{q}, \mathbf{P}$; we solve them for $\mathbf{P}$ in terms of $(\mathbf{q}, \mathbf{p})$. The second then delivers $\mathbf{Q}$ in terms of $(\mathbf{q}, \mathbf{P})$, hence in terms of $(\mathbf{q}, \mathbf{p})$. Checking that the result is always canonical is another careful application of the chain rule, which we omit.

In practice we usually already have a candidate for $\mathbf{P}$ in mind – typically some collection of first integrals – and we hunt for an S making the first equation hold. The second equation then *tells* us what the conjugate variable $\mathbf{Q}$ has to be. This is exactly how the Arnold–Liouville theorem will proceed.

Example 1.24

Take $S(\mathbf{q}, \mathbf{P}) = \mathbf{q} \cdot \mathbf{P}$. Then

$$\mathbf{p} = \frac{\partial S}{\partial \mathbf{q}} = \mathbf{P}, \qquad \mathbf{Q} = \frac{\partial S}{\partial \mathbf{P}} = \mathbf{q},$$

so S generates the identity transformation.

Example 1.25

In two dimensions, take $S(q, P) = qP + q^2$. Then

$$p = \frac{\partial S}{\partial q} = P + 2q, \qquad Q = \frac{\partial S}{\partial P} = q,$$

so the transformation is $(Q, P) = (q, p - 2q)$, or in matrix form

$$\begin{pmatrix} Q \\ P \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} q \\ p \end{pmatrix} = A \begin{pmatrix} q \\ p \end{pmatrix}.$$

We can verify directly that this is canonical:

$$AJA^T = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = J,$$

as promised.

§1.5 The Arnold–Liouville Theorem

We can now state the central result about integrable ODEs. Informally it says: if a Hamiltonian system is integrable, then there is a canonical transformation $(\mathbf{q}, \mathbf{p}) \mapsto (\boldsymbol{\phi}, \mathbf{I})$ after which the new Hamiltonian depends only on $\mathbf{I}$. If we can achieve this then Hamilton's equations become

$$\dot{\mathbf{I}} = -\frac{\partial \tilde{H}}{\partial \boldsymbol{\phi}} = 0, \qquad \dot{\boldsymbol{\phi}} = \frac{\partial \tilde{H}}{\partial \mathbf{I}},$$

so $\mathbf{I}(t) = \mathbf{I}_0$ is constant, and since the right-hand side of the second equation depends on $\mathbf{I}$ alone, $\dot{\boldsymbol{\phi}}$ is constant too. Hence

$$\boldsymbol{\phi}(t) = \boldsymbol{\phi}_0 + \boldsymbol{\Omega} t, \qquad \boldsymbol{\Omega} = \frac{\partial \tilde{H}}{\partial \mathbf{I}}(\mathbf{I}_0),$$

and we have solved the system completely. All the difficulty has been pushed into finding the transformation.

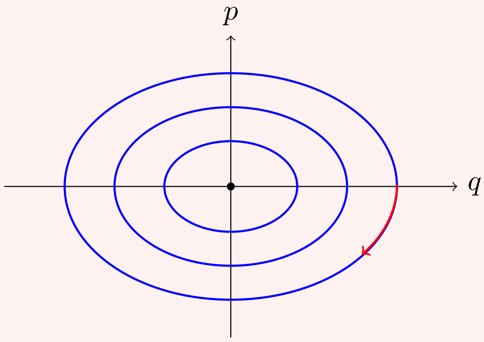
Before proving anything, let us see what the transformation looks like in the simplest possible example, where we can produce it by divine inspiration and then reverse-engineer where it came from.

Example 1.26 (Harmonic oscillator, first pass)

Take the harmonic oscillator on $M = \mathbb{R}^2$, with

$$H(q, p) = \tfrac{1}{2} p^2 + \tfrac{1}{2} \omega^2 q^2.$$

This is a two-dimensional system, so it is automatically integrable, with the single first integral $f_1 = H$. The level sets of H are ellipses:



Each of these is homeomorphic to a circle S^1, and this is not an accident – it is the first half of the Arnold–Liouville theorem in miniature.

Now introduce new coordinates (ϕ, I) by

$$q = \sqrt{\frac{2I}{\omega}} \sin\phi, \qquad p = \sqrt{2I\omega}\cos\phi.$$

One can check by computing the Jacobian that this is canonical. In these coordinates,

$$\tilde{H}(\phi, I) = \tfrac{1}{2} \cdot 2I\omega\cos^2\phi + \tfrac{1}{2}\omega^2 \cdot \frac{2I}{\omega}\sin^2\phi = \omega I.$$

There is no ϕ! So Hamilton's equations read

$$\dot{\phi} = \frac{\partial \tilde{H}}{\partial I} = \omega, \qquad \dot{I} = -\frac{\partial \tilde{H}}{\partial \phi} = 0,$$

giving $\phi(t) = \phi_0 + \omega t$ and $I(t) = I_0$. The oscillator goes round its ellipse at constant angular rate ω, which is of course the right answer.

Where could that transformation have come from? Here is a clue. Let us compute, for no obvious reason, the area enclosed by a level set of H, divided by 2π:

$$\begin{aligned}
\frac{1}{2\pi}\oint p \,\mathrm{d}q &= \frac{1}{2\pi}\int_0^{2\pi} p(\phi, I)\left(\frac{\partial q}{\partial \phi}\,\mathrm{d}\phi + \frac{\partial q}{\partial I}\,\mathrm{d}I\right) \\
&= \frac{1}{2\pi}\int_0^{2\pi} p(\phi, I)\frac{\partial q}{\partial \phi}\,\mathrm{d}\phi \\
&= \frac{1}{2\pi}\int_0^{2\pi} \sqrt{2I\omega}\cos\phi \cdot \sqrt{\frac{2I}{\omega}}\cos\phi\,\mathrm{d}\phi \\
&= \frac{2I}{2\pi}\int_0^{2\pi} \cos^2\phi\,\mathrm{d}\phi = I,
\end{aligned}$$

using $\mathrm{d}I = 0$ along a level set. So the new momentum I is nothing but the enclosed area over 2π – a quantity we could have computed knowing nothing whatsoever about the transformation.

There are two lessons here, and they are exactly the two halves of the theorem: the motion takes place on a circle, and the right momentum coordinate is $\frac{1}{2\pi}\oint p\,\mathrm{d}q$.

Theorem 1.27 (Arnold–Liouville)

Let (M, H) be an integrable $2n$-dimensional Hamiltonian system with independent, involutive first integrals $f_1, \ldots, f_n$, where $f_1 = H$. For $\mathbf{c} \in \mathbb{R}^n$ set

$$M_{\mathbf{c}} = \{(\mathbf{q}, \mathbf{p}) \in M : f_i(\mathbf{q}, \mathbf{p}) = c_i, \ i = 1, \ldots, n\}.$$

Then:

(i) $M_{\mathbf{c}}$ is a smooth n-dimensional surface in M, invariant under the flow. If $M_{\mathbf{c}}$ is compact and connected, it is diffeomorphic to the n-torus

$$T^n = \underbrace{S^1 \times \cdots \times S^1}_{n}.$$

(ii) If $M_{\mathbf{c}}$ is compact and connected, then locally there is a canonical transformation $(\mathbf{q}, \mathbf{p}) \mapsto (\boldsymbol{\phi}, \mathbf{I})$ to **action–angle coordinates**, in which the ϕ_k are angular coordinates on $M_{\mathbf{c}}$, the actions I_k are first integrals, and $\tilde{H}$ is independent of $\boldsymbol{\phi}$. Consequently

$$\dot{\mathbf{I}} = 0, \qquad \dot{\boldsymbol{\phi}} = \frac{\partial \tilde{H}}{\partial \mathbf{I}} = \text{constant}.$$

The first part is pure differential geometry, and the second is where the work is. We give the argument in outline; readers who have not met the preimage theorem or the orbit–stabiliser theorem in this setting may take the geometric steps on faith.

Proof sketch. Part (i). That $M_{\mathbf{c}}$ is a smooth n-dimensional surface is the preimage theorem (a consequence of the inverse function theorem from IB Analysis and Topology) applied to $\mathbf{f} = (f_1, \ldots, f_n) : M \to \mathbb{R}^n$. What makes it apply is precisely the independence of the f_i, which says that $\mathbf{c}$ is a regular value.

Next we show that $M_{\mathbf{c}}$ is a torus when compact and connected. Consider the Hamiltonian vector fields

$$\mathbf{V}_{f_i} = J \frac{\partial f_i}{\partial \mathbf{x}}.$$

We claim each is tangent to $M_{\mathbf{c}}$. It suffices to show that the derivative of each f_j along $\mathbf{V}_{f_i}$ vanishes, and indeed

$$\left(\mathbf{V}_{f_i} \cdot \frac{\partial}{\partial \mathbf{x}} \right) f_j = \frac{\partial f_j}{\partial \mathbf{x}} \cdot J \frac{\partial f_i}{\partial \mathbf{x}} = \{f_j, f_i\} = 0$$

by involutivity. So the flows g_i^t generated by the $\mathbf{V}_{f_i}$ map $M_{\mathbf{c}}$ into itself. Moreover these flows commute, since

$$[\mathbf{V}_{f_i}, \mathbf{V}_{f_j}] = -\mathbf{V}_{\{f_i, f_j\}} = -\mathbf{V}_0 = 0.$$

So writing $\mathbf{t} = (t_1, \ldots, t_n) \in \mathbb{R}^n$ and

$$g^{\mathbf{t}} = g_1^{t_1} g_2^{t_2} \cdots g_n^{t_n},$$

commutativity gives $g^{\mathbf{t}_1 + \mathbf{t}_2} = g^{\mathbf{t}_1} g^{\mathbf{t}_2}$, so we have an action of the group $(\mathbb{R}^n, +)$ on $M_{\mathbf{c}}$.

Fix $\mathbf{x} \in M_{\mathbf{c}}$ and let

$$\operatorname{Stab}(\mathbf{x}) = \{\mathbf{t} \in \mathbb{R}^n : g^{\mathbf{t}}\mathbf{x} = \mathbf{x}\}.$$

By the orbit–stabiliser theorem, $\mathbf{t} \mapsto g^{\mathbf{t}}\mathbf{x}$ descends to a bijection $\mathbb{R}^n/\operatorname{Stab}(\mathbf{x}) \to \operatorname{Orb}(\mathbf{x})$, and one checks that the orbit of $\mathbf{x}$ is the connected component of $\mathbf{x}$ in $M_{\mathbf{c}}$ – so if $M_{\mathbf{c}}$ is connected, the orbit is everything. Now $\operatorname{Stab}(\mathbf{x})$ is a subgroup of $\mathbb{R}^n$, and (provided the flows are non-degenerate) it is discrete, hence isomorphic to $\mathbb{Z}^k$ for some $0 \le k \le n$. Therefore

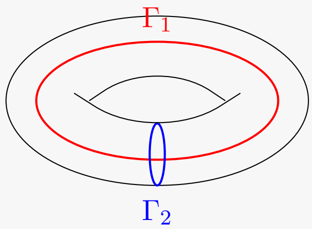
$$M_{\mathbf{c}} \cong \mathbb{R}^n/\mathbb{Z}^k \cong T^k \times \mathbb{R}^{n-k}.$$

If $M_{\mathbf{c}}$ is compact there can be no $\mathbb{R}$ factors, so $k = n$ and $M_{\mathbf{c}} \cong T^n$.

Part (ii). We construct the action–angle coordinates. For clarity of presentation we take $n = 2$; the general case is identical except that we need a higher-dimensional version of Green's theorem, which we do not have to hand.

It is trivially easy to find *some* coordinates $(\mathbf{Q}, \mathbf{P})$ with $\mathbf{P}$ constant along the flow and $\mathbf{Q}$ angular: just set $\mathbf{P} = \mathbf{c}$ and use the diffeomorphism $M_{\mathbf{c}} \cong T^n$ to get angles. The problem is that such a transformation will almost never be canonical. So we must be cleverer about the choice of $\mathbf{P}$.

On the torus T^n there are n topologically distinct 'basic' loops $\Gamma_1, \ldots, \Gamma_n$ – under the identification $M_{\mathbf{c}} \cong S^1 \times \cdots \times S^1$, the loop Γ_j runs once around the jth factor and is constant on the others. For $n = 2$ these are the two obvious circles:



We now define the **actions**

$$I_j = \frac{1}{2\pi} \oint_{\Gamma_j} \mathbf{p} \cdot \, \mathrm{d}\mathbf{q},$$

generalising the formula we found for the harmonic oscillator. The first thing to check is that this is well defined, since Γ_j is really a whole homotopy class of loops rather than a single one. So suppose Γ_2 and Γ_2' are two such loops, bounding between them an annular region A on the torus.

On $M_{\mathbf{c}}$ the equations $f_i(\mathbf{q}, \mathbf{p}) = c_i$ hold identically, and we assume we may invert them locally to write $\mathbf{p} = \mathbf{p}(\mathbf{q}, \mathbf{c})$; the condition for this is $\det(\partial f_i/\partial p_j) \neq 0$. Differentiating the identity $f_i(\mathbf{q}, \mathbf{p}(\mathbf{q}, \mathbf{c})) = c_i$ with respect to q_k gives

$$\frac{\partial f_i}{\partial q_k} + \frac{\partial f_i}{\partial p_\ell}\frac{\partial p_\ell}{\partial q_k} = 0$$

on $M_{\mathbf{c}}$. Now use involutivity: on $M_{\mathbf{c}}$,

$$0 = \{f_i, f_j\}$$

$$
\begin{aligned}
&= \frac{\partial f_i}{\partial q_k}\frac{\partial f_j}{\partial p_k} - \frac{\partial f_i}{\partial p_k}\frac{\partial f_j}{\partial q_k} \\
&= \left(-\frac{\partial f_i}{\partial p_\ell}\frac{\partial p_\ell}{\partial q_k}\right)\frac{\partial f_j}{\partial p_k} - \frac{\partial f_i}{\partial p_k}\left(-\frac{\partial f_j}{\partial p_\ell}\frac{\partial p_\ell}{\partial q_k}\right) \\
&= \frac{\partial f_i}{\partial p_k}\left(\frac{\partial p_\ell}{\partial q_k} - \frac{\partial p_k}{\partial q_\ell}\right)\frac{\partial f_j}{\partial p_\ell},
\end{aligned}
$$

where in the last line we relabelled dummy indices in the first term. Since the matrices $\partial f_i/\partial p_k$ are invertible, the matrix in the middle must vanish:

$$\frac{\partial p_\ell}{\partial q_k} = \frac{\partial p_k}{\partial q_\ell}.$$

For $n = 2$ the only content of this is $\partial p_1/\partial q_2 = \partial p_2/\partial q_1$. But then by Green's theorem

$$\left(\oint_{\Gamma_2} - \oint_{\Gamma_2'}\right)\mathbf{p}\cdot\, \mathrm{d}\mathbf{q} = \oint_{\partial A}\mathbf{p}\cdot\, \mathrm{d}\mathbf{q} = \iint_A \left(\frac{\partial p_2}{\partial q_1} - \frac{\partial p_1}{\partial q_2}\right)\, \mathrm{d}q_1\, \mathrm{d}q_2 = 0.$$

So I_j depends only on the homotopy class of Γ_j, and hence $\mathbf{I} = \mathbf{I}(\mathbf{c})$ is a function of $\mathbf{c}$ alone. These are our new momenta. We assume $\mathbf{I}(\mathbf{c})$ is invertible, so that we may also write $\mathbf{c} = \mathbf{c}(\mathbf{I})$.

It remains to produce the conjugate angles, and for this we use a generating function. Pick a base point $\mathbf{x}_0 \in M_\mathbf{c}$ and set

$$S(\mathbf{q}, \mathbf{I}) = \int_{\mathbf{x}_0}^{\mathbf{x}} \mathbf{p}(\mathbf{q}', \mathbf{c}(\mathbf{I}))\cdot\, \mathrm{d}\mathbf{q}',$$

where $\mathbf{x} = (\mathbf{q}, \mathbf{p}(\mathbf{q}, \mathbf{c}(\mathbf{I})))$ and the integral is along any path in $M_\mathbf{c}$. Is this well defined? If we deform the path slightly, sweeping out a region B, then S changes by $\oint_{\partial B}\mathbf{p}\cdot\, \mathrm{d}\mathbf{q}$, which vanishes by exactly the Green's theorem computation above. But we are on a torus, and there are paths that cannot be deformed into one another: a path may wind around one of the cycles Γ_j on its way. Adding such a loop changes S by $\oint_{\Gamma_j}\mathbf{p}\cdot\, \mathrm{d}\mathbf{q} = 2\pi I_j$. So the total ambiguity is

$$S(\mathbf{q}, \mathbf{I}) \mapsto S(\mathbf{q}, \mathbf{I}) + 2\pi m_j I_j, \qquad m_j \in \mathbb{Z},$$

and nothing worse. Differentiating with respect to $\mathbf{I}$, we conclude that

$$\boldsymbol{\phi} = \frac{\partial S}{\partial \mathbf{I}}$$

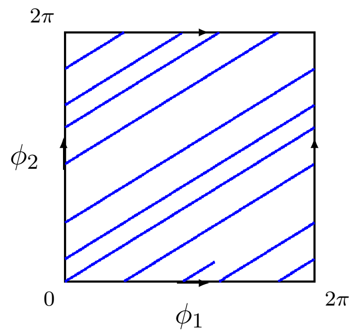
is well defined *modulo* 2π. These are the angles, and the ambiguity is exactly the ambiguity one expects of an angular coordinate.

Finally we should check that S really is a generating function in the sense of the previous subsection, i.e. that $\partial S/\partial\mathbf{q} = \mathbf{p}$. Writing $S = \int_{\mathbf{x}_0}^{\mathbf{x}}\mathbf{F}\cdot\, \mathrm{d}\mathbf{x}'$ with $\mathbf{F} = (\mathbf{p}, \mathbf{0})$, the fundamental theorem of calculus gives $\partial S/\partial\mathbf{x} = \mathbf{F}$, and in particular $\partial S/\partial\mathbf{q} = \mathbf{p}$ as required.

So S generates a canonical transformation $(\mathbf{q},\mathbf{p}) \mapsto (\boldsymbol{\phi},\mathbf{I})$. Since $\mathbf{I} = \mathbf{I}(\mathbf{c}) = \mathbf{I}(\mathbf{f}(\mathbf{x}))$ is a function of the first integrals alone, we have $\dot{\mathbf{I}} = 0$; and since $\dot{\mathbf{I}} = -\partial \tilde{H}/\partial \boldsymbol{\phi}$, the new Hamiltonian is independent of $\boldsymbol{\phi}$. Integrating up,

$$\boldsymbol{\phi}(t) = \boldsymbol{\phi}_0 + \boldsymbol{\Omega} t, \qquad \mathbf{I}(t) = \mathbf{I}_0, \qquad \boldsymbol{\Omega} = \frac{\partial \tilde{H}}{\partial \mathbf{I}}(\mathbf{I}_0). \qquad \square$$

The picture that emerges is a very appealing one. Phase space is foliated by invariant tori $M_{\mathbf{c}}$, one for each value of the conserved quantities, and on each torus the motion is uniform straight-line motion in the angle variables – a winding line, going round the jth cycle at frequency Ω_j. Drawing the torus in the usual way as a square with opposite edges identified, the trajectory is simply a straight line of slope Ω_2/Ω_1:



If the frequencies $\Omega_1, \ldots, \Omega_n$ are rationally related the trajectory closes up and the motion is periodic; if they are rationally independent the trajectory winds around forever, densely filling the torus, and the motion is called **quasi-periodic**.

§1.6 Action–Angle Variables

The proof above is constructive, and gives us a recipe. To integrate up an integrable Hamiltonian system:

(1) Identify the n cycles $\Gamma_1, \ldots, \Gamma_n$ on the invariant torus $M_{\mathbf{c}}$, and solve $f_i(\mathbf{q},\mathbf{p}) = c_i$ for $\mathbf{p} = \mathbf{p}(\mathbf{q},\mathbf{c})$.

(2) Compute the actions
$$I_j = \frac{1}{2\pi} \oint_{\Gamma_j} \mathbf{p} \cdot \, \mathrm{d}\mathbf{q},$$
and invert to obtain $\mathbf{c} = \mathbf{c}(\mathbf{I})$.

(3) Form the generating function
$$S(\mathbf{q},\mathbf{I}) = \int^{\mathbf{x}} \mathbf{p}(\mathbf{q}', \mathbf{c}(\mathbf{I})) \cdot \, \mathrm{d}\mathbf{q}'$$
and read off the angles $\boldsymbol{\phi} = \partial S/\partial \mathbf{I}$.

(4) Write $\tilde{H} = \tilde{H}(\mathbf{I})$ and read off the frequencies $\boldsymbol{\Omega} = \partial \tilde{H}/\partial \mathbf{I}$.

In practice step (4) is often all one wants: the frequencies of an integrable system can be extracted without ever computing the angles, since $\tilde{H}(\mathbf{I})$ is determined by step (2) alone.

Let us run the recipe on the harmonic oscillator, and see the transformation appear without any divine intervention.

Example 1.28 (Harmonic oscillator, second pass)
With $H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2$ we have

$$M_c = \left\{(q,p) : \tfrac{1}{2}p^2 + \tfrac{1}{2}\omega^2 q^2 = c\right\},$$

an ellipse, which is indeed diffeomorphic to $T^1 = S^1$ as the theorem promises. There is only one cycle, the ellipse itself. Solving for p,

$$p = p(q,c) = \pm\sqrt{2c - \omega^2 q^2}.$$

The action is the enclosed area over 2π. The ellipse $\frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2 = c$ has semi-axes $\sqrt{2c}/\omega$ and $\sqrt{2c}$, so its area is $\pi \cdot \frac{\sqrt{2c}}{\omega} \cdot \sqrt{2c} = 2\pi c/\omega$ and

$$I = \frac{1}{2\pi}\oint p \,\mathrm{d}q = \frac{c}{\omega}.$$

Inverting, $c = c(I) = \omega I$, so immediately

$$\tilde{H} = c = \omega I, \qquad \Omega = \frac{\mathrm{d}\tilde{H}}{\mathrm{d}I} = \omega.$$

The frequency is ω, independent of amplitude – the defining feature of the harmonic oscillator.

For the angle, take the base point to be $q = 0$ with $p > 0$. Then

$$S(q,I) = \int_0^q \sqrt{2\omega I - \omega^2 q'^2}\,\mathrm{d}q',$$

and differentiating under the integral sign,

$$\phi = \frac{\partial S}{\partial I} = \int_0^q \frac{\omega\,\mathrm{d}q'}{\sqrt{2\omega I - \omega^2 q'^2}} = \int_0^q \frac{\mathrm{d}q'}{\sqrt{2I/\omega - q'^2}} = \sin^{-1}\left(\sqrt{\frac{\omega}{2I}}\,q\right).$$

Sure enough, this is well defined only up to multiples of 2π. Inverting,

$$q = \sqrt{\frac{2I}{\omega}}\sin\phi, \qquad p = \frac{\partial S}{\partial q} = \sqrt{2\omega I - \omega^2 q^2} = \sqrt{2I\omega}\cos\phi,$$

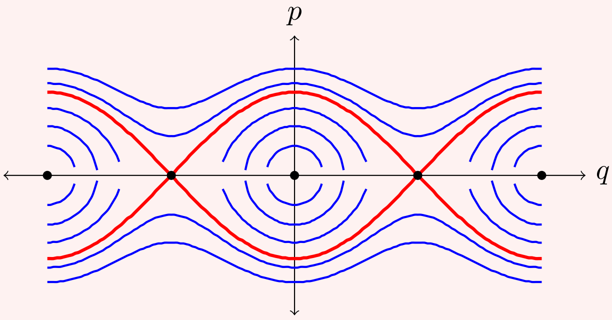
which is exactly the transformation we pulled out of thin air earlier.

The next example is the one to remember, because it is the simplest system in which the frequency actually depends on the action.

Example 1.29 (The simple pendulum)
A pendulum of unit mass and unit length in unit gravity has Hamiltonian

$$H(q,p) = \tfrac{1}{2}p^2 - \cos q,$$

where q is the angle from the downward vertical. The phase portrait is the following, with the equilibria marked: centres at $q = 0, \pm 2\pi, \ldots$ (the pendulum hanging down) and saddles at $q = \pm\pi$ (the pendulum balanced upright).



There are three qualitatively different kinds of motion, separated by the thick red curve.

(i) For $-1 < E < 1$ the level set $\{H = E\}$ is a closed curve encircling the origin. This is **libration**: the pendulum swings back and forth about the bottom, with q bounded by the turning points $\pm q_0$ where $\cos q_0 = -E$.

(ii) For $E > 1$ the level set consists of two curves, one with $p > 0$ throughout and one with $p < 0$. This is **rotation**: the pendulum has enough energy to go over the top, and q increases (or decreases) without bound.

(iii) At $E = 1$ we have the **separatrix**, $p = \pm 2\cos(q/2)$, on which the pendulum takes infinite time to creep up to the inverted position $q = \pi$.

Note that the phase space here is really a cylinder rather than a plane, since q and $q + 2\pi$ describe the same configuration; on the cylinder the libration curves are circles, and so are the rotation curves, but the two families of circles are not homotopic to one another.

Consider the librating region. Solving for p,

$$p = \pm\sqrt{2(E + \cos q)},$$

and the action is

$$I(E) = \frac{1}{2\pi}\oint p \, \mathrm{d}q = \frac{1}{\pi}\int_{-q_0}^{q_0} \sqrt{2(E + \cos q)} \, \mathrm{d}q,$$

where $q_0 = \cos^{-1}(-E)$ and the factor of two from the upper and lower halves of the curve has been absorbed. Using $\cos q = 1 - 2\sin^2(q/2)$ and writing $k^2 = (1+E)/2 \in (0, 1)$, this becomes

$$I(E) = \frac{4}{\pi}\int_0^{q_0} \sqrt{k^2 - \sin^2(q/2)} \, \mathrm{d}q = \frac{8}{\pi}\left[\mathcal{E}(k) - (1 - k^2)\mathcal{K}(k)\right],$$

where $\mathcal{K}$ and $\mathcal{E}$ are the complete elliptic integrals of the first and second kind. This is not an elementary function, and that is the point: the pendulum is integrable, but its action–angle variables involve elliptic functions. The frequency is

$$\Omega(I) = \frac{\mathrm{d}E}{\mathrm{d}I} = \left(\frac{\mathrm{d}I}{\mathrm{d}E}\right)^{-1} = \frac{\pi}{2\mathcal{K}(k)},$$

using $\mathrm{d}I/\,\mathrm{d}E = \frac{2}{\pi}\mathcal{K}(k)$, which follows from differentiating the integral for I under the integral sign.

It is worth sanity-checking the two limits. As $E \to -1^+$ we have $k \to 0$, and $\mathcal{K}(0) = \pi/2$, so $\Omega \to 1$: small oscillations have unit frequency, in agreement with linearising $\ddot{q} = -\sin q \approx -q$. As $E \to 1^-$ we have $k \to 1$ and $\mathcal{K}(k) \to \infty$, so $\Omega \to 0$: the period diverges as we approach the separatrix, which is the statement that the pendulum takes forever to reach the top. The pendulum is thus the standard example of an integrable system whose frequency depends on its amplitude.

Our final example has two degrees of freedom, so that we get an honest two-torus and two genuinely different cycles.

Example 1.30 (A particle in a central potential)

Consider a particle of unit mass moving in a plane under a central potential $U(r)$. Using polar coordinates (r, θ), the Hamiltonian is

$$H = \frac{1}{2}\left(p_r^2 + \frac{p_\theta^2}{r^2}\right) + U(r),$$

with the canonical pairs (r, p_r) and (θ, p_θ). This is a four-dimensional phase space, so we need two first integrals in involution, and we have them: $f_1 = H$ and $f_2 = p_\theta$, the angular momentum. They are in involution because H does not depend on θ:

$$\{p_\theta, H\} = -\frac{\partial H}{\partial \theta} = 0.$$

So the system is integrable. Fix $\mathbf{c} = (E, L)$. The invariant surface $M_{\mathbf{c}}$ is described by

$$p_\theta = L, \qquad p_r = \pm\sqrt{2(E - U(r)) - \frac{L^2}{r^2}},$$

and provided the effective potential $U_{\text{eff}}(r) = U(r) + L^2/2r^2$ has a well, so that r oscillates between turning points r_- and r_+, this really is a torus: θ runs around a circle, and (r, p_r) traces a closed curve in its own plane.

The two cycles are then evident. Taking Γ_1 to be the loop on which θ increases by 2π at fixed r, and Γ_2 the loop on which (r, p_r) traverses its closed curve at fixed θ, we get

$$I_1 = \frac{1}{2\pi}\oint_{\Gamma_1} p_\theta \, \mathrm{d}\theta = L, \qquad I_2 = \frac{1}{2\pi}\oint_{\Gamma_2} p_r \, \mathrm{d}r = \frac{1}{\pi}\int_{r_-}^{r_+} \sqrt{2(E - U(r)) - \frac{L^2}{r^2}} \, \mathrm{d}r.$$

The first action is just the angular momentum, as one would hope. The second is the **radial action**, and computing it gives E as a function of (I_1, I_2) and hence both frequencies at once.

For the Kepler potential $U(r) = -1/r$ this integral can be done by contour methods, with the answer

$$I_2 = -I_1 + \frac{1}{\sqrt{-2E}}, \qquad \text{i.e.} \qquad \tilde{H} = E = -\frac{1}{2(I_1 + I_2)^2}.$$

The remarkable thing here is that $\tilde{H}$ depends on I_1 and I_2 only through the combination $I_1 + I_2$, so that

$$\Omega_1 = \frac{\partial \tilde{H}}{\partial I_1} = \frac{\partial \tilde{H}}{\partial I_2} = \Omega_2 = \frac{1}{(I_1 + I_2)^3}.$$

The two frequencies are equal, so the winding line on the torus closes up after one circuit: Kepler orbits are closed curves, which is the statement that planetary orbits are ellipses that do not precess. This **degeneracy** is special to the $1/r$ potential (and to the harmonic oscillator), and its breaking by relativistic corrections is what produces the precession of the perihelion of Mercury.

This is a good point to pause and take stock. We have a complete and satisfying theory: a Hamiltonian system with enough commuting conserved quantities lives on invariant tori, and on those tori the motion is trivial. For the rest of the course we ask whether anything like this survives when we pass to partial differential equations. The answer, remarkably, is yes – but almost every ingredient will have to be reinvented.

§2 Integrable PDEs and Solitons

For the rest of the course we turn to partial differential equations. It is helpful to think of a PDE as an infinite-dimensional dynamical system: instead of finitely many coordinates $x_i(t)$ we have a whole function $u(x,t)$, with the spatial variable x playing the role of the index i.

What should we expect of an integrable PDE? If a $2n$-dimensional ODE is integrable when it has n first integrals, and our phase space is now infinite-dimensional, then – since half of infinity is still infinity – we should expect an integrable PDE to possess *infinitely many* conserved quantities. We should also expect some magic transformation, analogous to the passage to action–angle variables, which turns the equation into something trivial. Both of these expectations turn out to be correct. What is much less obvious is how to find the transformation, and we will spend the whole of Section 3 constructing it for one particular equation.

We will not attempt to define integrability for PDEs. Nobody has managed to do so in a way that is both precise and useful, and we will be content to work with examples.

§2.1 Dispersion and Nonlinearity

The equation at the centre of this course is the Korteweg–de Vries equation

$$u_t + u_{xxx} - 6uu_x = 0.$$

Before meeting it properly, it pays to understand the two terms u_{xxx} and $-6uu_x$ separately, because each on its own does something violent to a wave, and the whole point of KdV is that the two violences cancel.

Example 2.1 (Dispersion)

Consider the *linear* equation

$$u_t + u_{xxx} = 0.$$

Being linear and constant-coefficient, it is solved by Fourier methods (IB Methods). Plane wave modes $u = e^{i(kx-\omega t)}$ solve it provided ω satisfies the **dispersion relation**

$$\omega = \omega(k) = -k^3.$$

The general solution is then a superposition,

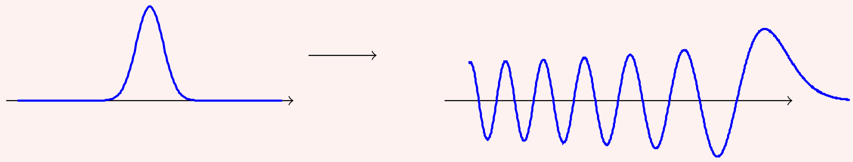
$$u(x,t) = \int_{-\infty}^{\infty} A(k) e^{i(kx-\omega(k)t)}\, \mathrm{d}k,$$

with $A(k) = \hat{u}_0(k)/2\pi$ determined by the initial data.

Writing a single mode as

$$u(x,t) = \exp\left(ik\left(x - \frac{\omega(k)}{k}t\right)\right),$$

we see that it travels with phase speed $\omega/k = -k^2$. The crucial point is that this depends on k: modes of different wavelength travel at different speeds. So an initial profile, which is a superposition of many modes, does not translate rigidly – it spreads out, with the short wavelengths racing off in one direction and the long wavelengths lagging behind. This is **dispersion**, and it is the work of the u_{xxx} term. A localised initial bump decays into an oscillatory wave train:



Had we written $u_t + u_x = 0$ instead, the dispersion relation would be $\omega = k$, the phase speed would be 1 for every mode, and the initial profile would translate rigidly. It is the *third* derivative that does the damage.

Example 2.2 (Wave breaking)

Now consider the *nonlinear* equation

$$u_t - 6uu_x = 0.$$

This looks worse, being nonlinear, but in fact it can be solved exactly for any initial data $u(x,0) = f(x)$ by the method of characteristics. The characteristics are the curves along which u is constant, and since the equation says u is transported at speed $-6u$, they are straight lines. The solution is given implicitly by

$$u(x,t) = f(\xi), \qquad \xi = x - 6tf(\xi).$$

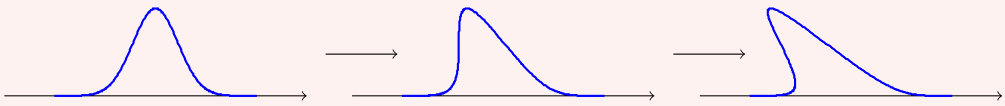
This is fine as long as the implicit relation can be solved for ξ. Let us see when it fails. Differentiating,

$$u_x = f'(\xi)\frac{\partial \xi}{\partial x}, \qquad \frac{\partial \xi}{\partial x} = 1 - 6tf'(\xi)\frac{\partial \xi}{\partial x},$$

so

$$\frac{\partial \xi}{\partial x} = \frac{1}{1 + 6tf'(\xi)}, \qquad u_x = \frac{f'(\xi)}{1 + 6tf'(\xi)}.$$

Whenever f' takes a negative value somewhere, this blows up at the finite time $t = -1/(6f'(\xi))$: the slope becomes infinite, and after that the implicit formula would give a multi-valued 'solution'.



This is **wave breaking**, and it is what the nonlinear term $-6uu_x$ does on its own: taller parts of the wave move faster, overtake the parts in front of them, and the wave topples over.

So dispersion spreads a wave out and destroys it, and nonlinearity steepens a wave and destroys it. What happens if we include both?

§2.2 The Korteweg–de Vries Equation

Definition 2.3 (KdV equation)

The **Korteweg–de Vries equation**, or **KdV equation**, is

$$u_t + u_{xxx} - 6uu_x = 0.$$

The equation has a distinguished history. In 1834 the Scottish engineer John Scott Russell was watching a boat being drawn along a canal by a pair of horses. When the boat stopped suddenly, the water it had been pushing rolled forward as a single smooth heap, and Russell – to his eternal credit – got on his horse and chased it for a mile or two along the canal, watching it travel without change of shape or speed. He called it the 'wave of translation'. It took until 1895 for Korteweg and de Vries to write down the equation governing it.

Let us see, at least at the level of scaling, where the equation comes from.

Example 2.4 (Shallow water waves)

Consider small-amplitude surface waves on a layer of water of undisturbed depth h, and let $u(x,t)$ denote the surface elevation. Linear water wave theory gives the exact dispersion relation

$$\omega^2 = gk \tanh(kh).$$

For long waves, $kh \ll 1$, we expand $\tanh z = z - z^3/3 + O(z^5)$ to get

$$\omega^2 = gk\left(kh - \frac{(kh)^3}{3} + \cdots\right) = c^2k^2\left(1 - \frac{(kh)^2}{3} + \cdots\right), \qquad c = \sqrt{gh},$$

so that for right-moving waves

$$\omega = ck\left(1 - \frac{k^2h^2}{6} + \cdots\right) = ck - \frac{ch^2}{6}k^3 + \cdots.$$

A wave packet whose modes obey $\omega = ck - \beta k^3$ satisfies the linear PDE

$$u_t + cu_x + \beta u_{xxx} = 0, \qquad \beta = \frac{ch^2}{6},$$

since $\partial_t \to -i\omega$ and $\partial_x \to ik$.

Now put back the leading nonlinearity. A finite-amplitude long wave rides on water of local depth $h + u$, so it travels a little faster where it is taller; carrying out the expansion properly gives a correction to the propagation speed of $\frac{3}{2}cu/h$, and hence a term $\gamma u u_x$ with $\gamma = 3c/2h$. Putting both effects together,

$$u_t + cu_x + \gamma u u_x + \beta u_{xxx} = 0.$$

Finally we normalise. Passing to a frame moving at speed c kills the cu_x term, and setting $u = Av(y, s)$ with $y = x - ct$ and $s = t/\beta$ gives

$$v_s + \frac{\gamma A}{\beta} v v_y + v_{yyy} = 0.$$

Choosing $A = -6\beta/\gamma$ produces

$$v_s + v_{yyy} - 6 v v_y = 0,$$

which is KdV. So the equation is not some arbitrary combination of terms: it is what one gets from *any* weakly nonlinear, weakly dispersive, unidirectional wave system at leading order. This universality is why KdV turns up in plasma physics, in nonlinear optics, in blood flow and in lattice dynamics as well as in canals.

Some structural remarks are worth making before we solve anything. The KdV equation is invariant under

(i) translations $x \mapsto x + a$, $t \mapsto t + b$;

(ii) the scaling $u \mapsto \lambda^2 u$, $x \mapsto \lambda^{-1} x$, $t \mapsto \lambda^{-3} t$;

(iii) the Galilean boost $u(x,t) \mapsto u(x - 6ct, t) + c$ for constant c.

The third is easy to miss and will matter later: adding a constant to u is the same as moving to a uniformly translating frame. The scaling symmetry tells us that there is really only a one-parameter family of solitary waves, all obtainable from one another by rescaling.

§2.3 Travelling Waves and the One-Soliton Solution

We now look for travelling wave solutions of KdV, and we shall find Russell's wave of translation. Let

$$u(x,t) = f(\xi), \qquad \xi = x - ct,$$

for some speed c, and impose the boundary conditions $f, f', f'' \to 0$ as $|\xi| \to \infty$ (we want a localised wave on still water). Substituting,

$$-cf' + f''' - 6ff' = 0.$$

Every term is a derivative, so we can integrate once:

$$-cf + f'' - 3f^2 = A$$

for a constant A, and letting $\xi \to \infty$ forces $A = 0$. Now multiply by the integrating factor f':

$$-cff' + f''f' - 3f^2 f' = 0,$$

and integrate again, noting that $f''f' = \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}\xi}(f')^2$:

$$-\frac{c}{2}f^2 + \frac{1}{2}(f')^2 - f^3 = B,$$

and again $B = 0$ by the boundary conditions. Rearranging,

$$(f')^2 = 2f^3 + cf^2 = f^2(2f + c). \tag{$\dagger$}$$

This is a first-order ODE which we could grind through by separation of variables. It is quicker to guess. Since $(f')^2 \geq 0$ we need $2f + c \geq 0$ wherever $f \neq 0$; if we look for a solution with $f < 0$ (which is the natural sign for KdV in this normalisation) this forces $c > 0$ and $-c/2 \leq f \leq 0$. A function which decays at both ends and has a minimum of $-c/2$ suggests sech^2, so we try

$$f(\xi) = -\frac{c}{2}\operatorname{sech}^2(\alpha\xi)$$

and see what α must be. Using $\frac{\mathrm{d}}{\mathrm{d}\theta}\operatorname{sech}\theta = -\operatorname{sech}\theta\tanh\theta$,

$$f'(\xi) = c\alpha\operatorname{sech}^2(\alpha\xi)\tanh(\alpha\xi), \qquad (f')^2 = c^2\alpha^2\operatorname{sech}^4\tanh^2.$$

On the other side of ($\dagger$),

$$f^2(2f + c) = \frac{c^2}{4}\operatorname{sech}^4 \cdot c\left(1 - \operatorname{sech}^2\right) = \frac{c^3}{4}\operatorname{sech}^4\tanh^2,$$

using $1 - \operatorname{sech}^2 = \tanh^2$. So ($\dagger$) holds precisely when $c^2\alpha^2 = c^3/4$, that is $\alpha = \frac{1}{2}\sqrt{c}$. We have found our solution.

Proposition 2.5 (One-soliton solution)
For any $c > 0$, the function

$$u(x,t) = -\frac{c}{2}\operatorname{sech}^2\left(\frac{\sqrt{c}}{2}(x - ct - x_0)\right)$$

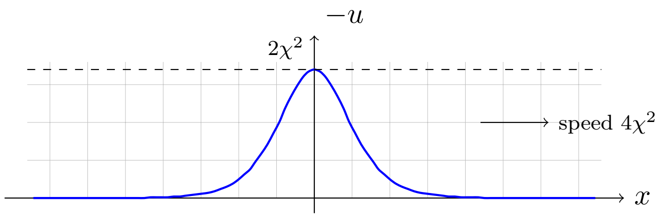
solves the KdV equation. Writing $c = 4\chi^2$ with $\chi > 0$, this takes the standard form

$$u(x,t) = -2\chi^2\operatorname{sech}^2\left(\chi(x - 4\chi^2 t - x_0)\right).$$

We call this a **soliton**. Look at what it says: the amplitude is $2\chi^2$, the speed is $4\chi^2$, and the width is $1/\chi$. So

taller solitons are faster and narrower.

This single fact is responsible for almost everything interesting that follows.



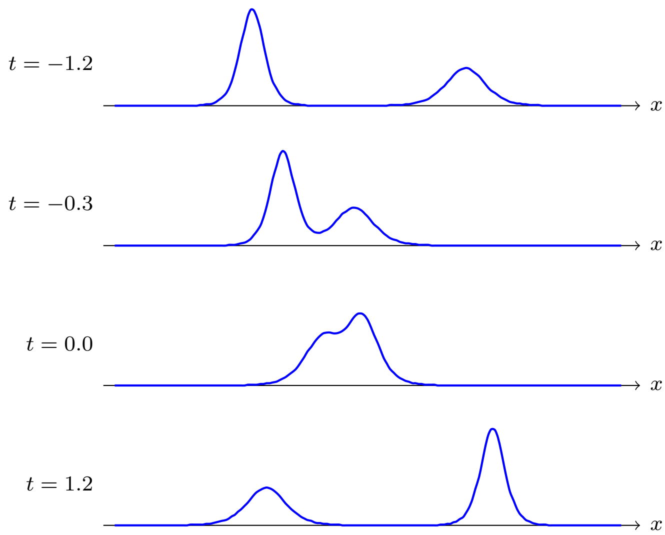
The balance is delicate. The dispersive term wants to spread the wave out; the nonlinear term wants to steepen its leading edge. For this particular profile, at this particular relation between height and speed, the two exactly cancel and the wave propagates unchanged forever.

Remark. The parameter χ is not a typographical accident. When we come to the inverse scattering transform we will find that $-\chi^2$ is precisely the eigenvalue of the Schrödinger operator $-\partial_x^2 + u$ associated with this potential. A one-soliton solution is the potential with exactly one bound state.

§2.4 Two Solitons, and Why Solitons Are Particles

Now for the interesting question. Suppose we start with two solitons, well separated, the taller one behind. Since the taller one is faster, it will catch up with the shorter one, and at some point they will overlap. The KdV equation is nonlinear, so we cannot simply add the two solutions; and we have every reason to expect that the collision will be messy, generating radiation and destroying both waves.

Here is what actually happens. (We will derive the exact formula in Section 3; for now we simply plot it, with $\chi_1 = 1$ and $\chi_2 = 1.6$. The window follows the interaction – both humps are in fact moving steadily to the right, and we plot $-u$ so that the solitons point upwards.)



The two solitons pass straight through one another and re-emerge with exactly their original shapes and speeds. Nothing is radiated away, nothing is lost. The only trace of the interaction is a **phase shift**: the taller soliton ends up slightly ahead of where it would have been, and the shorter one slightly behind. One can show that these shifts are

$$\Delta_2 = +\frac{1}{\chi_2} \log \frac{\chi_2 + \chi_1}{\chi_2 - \chi_1}, \qquad \Delta_1 = -\frac{1}{\chi_1} \log \frac{\chi_2 + \chi_1}{\chi_2 - \chi_1},$$

for the taller and shorter solitons respectively, so the taller one gains and the shorter one loses.

This is extraordinary behaviour for a nonlinear PDE, and it is the reason for the name: a 'soliton' is a solitary wave which behaves like a *particle* in collisions. The phenomenon was discovered numerically by Zabusky and Kruskal in 1965, who were simulating KdV as a continuum model for the Fermi–Pasta–Ulam problem and were startled to find their pulses surviving collisions intact. It was their work that started the modern theory of integrable systems.

Note also that in the middle panel, at the moment of collision, the combined profile is not the sum of the two solitons, nor even taller than the larger one – the interaction is genuinely nonlinear. Superposition is violated at every instant; it is only the asymptotics that are clean.

§2.5 Conservation Laws

If solitons survive collisions, something must be preventing energy from leaking into other modes. The something is an infinite family of conservation laws. Let us find the first few by hand.

We will always assume that u and its derivatives decay rapidly as $|x| \to \infty$, so that we may integrate by parts freely and discard boundary terms.

Definition 2.6 (Conservation law)
A **conservation law** for an evolution equation is a relation of the form

$$\frac{\partial T}{\partial t} + \frac{\partial X}{\partial x} = 0,$$

where the **density** T and the **flux** X are functions of x, t, u and the derivatives of u. If such a relation holds for all solutions, then

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_{-\infty}^{\infty} T \, \mathrm{d}x = -\Big[X\Big]_{-\infty}^{\infty} = 0,$$

so $\int T \, \mathrm{d}x$ is a first integral.

Proposition 2.7
The KdV equation possesses the conservation laws

$$\begin{aligned} \partial_t(u) + \partial_x\left(u_{xx} - 3u^2\right) &= 0, \\ \partial_t(u^2) + \partial_x\left(2uu_{xx} - u_x^2 - 4u^3\right) &= 0, \\ \partial_t\left(u^3 + \tfrac{1}{2}u_x^2\right) + \partial_x\left(\cdots\right) &= 0. \end{aligned}$$

Hence

$$I_1 = \int u \, \mathrm{d}x, \qquad I_2 = \int u^2 \, \mathrm{d}x, \qquad I_3 = \int \left(u^3 + \tfrac{1}{2}u_x^2\right) \, \mathrm{d}x$$

are first integrals of KdV.

Proof. The first is just KdV rewritten: $u_t = -u_{xxx} + 6uu_x = -\partial_x(u_{xx} - 3u^2)$.

For the second, multiply KdV by $2u$:

$$(u^2)_t = 2uu_t = 2u(-u_{xxx} + 6uu_x) = -2uu_{xxx} + 12u^2u_x.$$

We must write the right-hand side as $-\partial_x X$. Observe that

$$\partial_x(2uu_{xx} - u_x^2 - 4u^3) = 2u_xu_{xx} + 2uu_{xxx} - 2u_xu_{xx} - 12u^2u_x = 2uu_{xxx} - 12u^2u_x,$$

which is exactly $-(u^2)_t$, as required.

For the third we argue slightly differently, since guessing the flux is tedious. Set $I_3 = \int(u^3 + \frac{1}{2}u_x^2)\,\mathrm{d}x$ and differentiate:

$$\begin{aligned}\frac{\mathrm{d}I_3}{\mathrm{d}t} &= \int \left(3u^2u_t + u_xu_{xt}\right)\,\mathrm{d}x \\ &= \int \left(3u^2 - u_{xx}\right) u_t\,\mathrm{d}x,\end{aligned}$$

integrating the second term by parts. Now substitute $u_t = \partial_x(3u^2 - u_{xx})$ from the first conservation law, and write $w = 3u^2 - u_{xx}$. Then

$$\frac{\mathrm{d}I_3}{\mathrm{d}t} = \int w\,w_x\,\mathrm{d}x = \frac{1}{2}\int \partial_x(w^2)\,\mathrm{d}x = 0. \qquad \square$$

Physically, I_1 is the mass, I_2 is the momentum, and I_3 will turn out to be the Hamiltonian for KdV, which we may reasonably call the energy. The pattern continues: there is a conserved density involving u^4, another involving u^5, and so on forever. Finding them one at a time by inspection quickly becomes hopeless, and in Section 3 we will produce the whole infinite family in one stroke using the scattering problem.

Remark. Not every conservation law is interesting. For instance $\partial_t(u_x) + \partial_x(u_{xx} - 3u^2)_x = 0$ gives the first integral $\int u_x\,\mathrm{d}x = 0$, which is true but vacuous, since it holds for any decaying function whatsoever. We will meet a systematic supply of such trivial laws later, and we will have to learn to discard them.

§2.6 The Miura Transform

We finish this section with a piece of magic whose significance will only become clear in Section 3. Alongside KdV sits its close relative.

Definition 2.8 (mKdV)

The **modified KdV equation** is

$$v_t + v_{xxx} - 6v^2v_x = 0.$$

The two equations look similar but are genuinely different (mKdV, for instance, is invariant under $v \mapsto -v$, which KdV is not). Nevertheless, they are related.

Proposition 2.9 (Miura transform)

Define the **Miura transform**

$$u = v^2 + v_x.$$

Then

$$u_t + u_{xxx} - 6uu_x = (2v + \partial_x)\left(v_t + v_{xxx} - 6v^2v_x\right).$$

In particular, if v solves mKdV then $u = v^2 + v_x$ solves KdV.

Proof. This is a direct computation. From $u = v^2 + v_x$ we get

$$\begin{aligned} u_t &= 2vv_t + v_{xt}, \\ u_x &= 2vv_x + v_{xx}, \\ u_{xxx} &= 6v_xv_{xx} + 2vv_{xxx} + v_{xxxx}, \\ 6uu_x &= 6(v^2 + v_x)(2vv_x + v_{xx}) = 12v^3v_x + 6v^2v_{xx} + 12vv_x^2 + 6v_xv_{xx}. \end{aligned}$$

Adding up,

$$\begin{aligned} u_t + u_{xxx} - 6uu_x &= 2vv_t + v_{xt} + 6v_xv_{xx} + 2vv_{xxx} + v_{xxxx} \\ &\quad - 12v^3v_x - 6v^2v_{xx} - 12vv_x^2 - 6v_xv_{xx} \\ &= 2v\left(v_t + v_{xxx} - 6v^2v_x\right) + \left(v_{xt} + v_{xxxx} - 6v^2v_{xx} - 12vv_x^2\right). \end{aligned}$$

On the other hand, writing $M = v_t + v_{xxx} - 6v^2v_x$,

$$\partial_x M = v_{xt} + v_{xxxx} - 6\left(2vv_x \cdot v_x + v^2v_{xx}\right) = v_{xt} + v_{xxxx} - 12vv_x^2 - 6v^2v_{xx},$$

which is exactly the bracketed expression above. So the two sides agree. □

This is already useful – it lets us manufacture solutions of KdV from solutions of mKdV. But the reason it is a landmark result is the following observation. The relation

$$u = v^2 + v_x$$

is a **Riccati equation** for v, and the standard trick for Riccati equations is to substitute

$$v = \frac{\psi_x}{\psi}.$$

Then

$$v_x = \frac{\psi_{xx}}{\psi} - \frac{\psi_x^2}{\psi^2}, \qquad v^2 = \frac{\psi_x^2}{\psi^2},$$

so $u = \psi_{xx}/\psi$, that is

$$-\psi_{xx} + u\psi = 0.$$

Better still, KdV has the Galilean symmetry noted above, so if u solves KdV then so does $u + \lambda$ suitably translated; applying the Miura transform to the shifted equation gives

$$-\psi_{xx} + u\psi = \lambda\psi.$$

Out of nowhere, the time-independent Schrödinger equation has appeared, with the solution of KdV playing the role of the potential. Gardner, Greene, Kruskal and Miura took this hint seriously in 1967, and what came out of it is the subject of the next section.

§3 The Inverse Scattering Transform

We now come to the heart of the course. In IB Methods we learned to solve linear PDEs with the Fourier transform: to solve

$$u_t + u_{xxx} = 0,$$

we transform in x to get $\hat{u}_t - ik^3\hat{u} = 0$, which is a trivial ODE in t for each k; we solve it, and transform back.

$$\begin{tikzcd}
u_0(x) \arrow[rr, "\text{Fourier transform}"] \arrow[d, dashed, "u_t + u_{xxx} = 0"'] & & \hat{u}_0(k) \arrow[d, "\hat{u}_t - ik^3\hat{u} = 0"] \\
u(x,t) & & \hat{u}(k,t) = \hat{u}_0(k)e^{ik^3 t} \arrow[ll, "\text{inverse Fourier transform}"]
\end{tikzcd}$$

The whole scheme rests on the fact that the transform diagonalises ∂_x, converting a differential equation into an algebraic one.

The inverse scattering transform follows exactly the same three-step pattern, but with a much cleverer transform. Given $u(x,t)$, for each fixed t we regard u as the potential in a Schrödinger operator and compute its *scattering data*. The KdV equation then dictates how the scattering data evolves, and – astonishingly – the evolution is linear and explicit. Finally we invert the transform to recover u.

The remarkable thing is that every step of this process is linear, and yet the equation we are solving is not.

We build the machine in pieces: first the forward scattering problem (Section 3.1), then the inverse problem (Section 3.2), then the mechanism – Lax pairs – which links the evolution of u to the evolution of the scattering data (Sections 3.3–3.4), and finally the payoff (Sections 3.5–3.6).

§3.1 The Forward Scattering Problem

Throughout this section L denotes the Schrödinger operator

$$L = -\frac{\partial^2}{\partial x^2} + u(x),$$

where the potential u has compact support, so that $u = 0$ for $|x|$ sufficiently large. What we actually need is only that u decays fast enough at infinity, but assuming compact support saves us from having to be careful, and costs nothing.

For a given u, the **forward scattering problem** is to find the eigenvalues and eigenfunctions of L, that is, to solve

$$L\psi = \lambda\psi.$$

The *inverse* problem, which we address next, is to reconstruct u from that data.

There are two regimes to consider, corresponding to $\lambda > 0$ and $\lambda < 0$.

§3.1.1 The Continuous Spectrum

Take $\lambda = k^2 > 0$ with k real. Since u vanishes for large $|x|$, any solution satisfies

$$\psi_{xx} + k^2\psi = 0$$

out there, so is asymptotically a combination of $e^{\pm ikx}$. We single out a particular solution $\varphi = \varphi(x,k)$ by the boundary condition

$$\varphi(x,k) = e^{-ikx} \quad \text{as } x \to -\infty,$$

and then there must exist coefficients $a(k)$ and $b(k)$ with

$$\varphi(x,k) = a(k)e^{-ikx} + b(k)e^{ikx} \quad \text{as } x \to +\infty.$$

Definition 3.1 (Reflection and transmission coefficients)
With φ as above, set

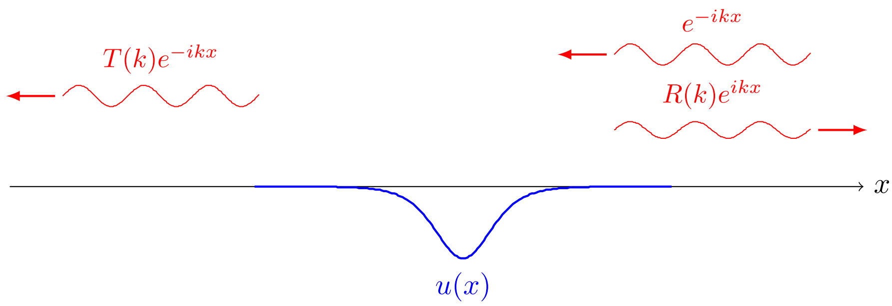
$$\Phi(x,k) = \frac{\varphi(x,k)}{a(k)}, \qquad R(k) = \frac{b(k)}{a(k)}, \qquad T(k) = \frac{1}{a(k)}.$$

We call R the **reflection coefficient** and T the **transmission coefficient**.

By construction,

$$\Phi(x,k) = \begin{cases} T(k)e^{-ikx} & x \to -\infty, \\ e^{-ikx} + R(k)e^{ikx} & x \to +\infty. \end{cases}$$

The physical reading is exactly the one from IB Quantum Mechanics. Thinking of e^{-ikx} as a leftward-moving wave and e^{ikx} as a rightward-moving one, we have a unit-amplitude wave incident from the right, of which a portion $R(k)e^{ikx}$ is reflected back to the right and a portion $T(k)e^{-ikx}$ is transmitted through to the left.



Conservation of probability flux (equivalently, constancy of the Wronskian) gives

$$|R(k)|^2 + |T(k)|^2 = 1,$$

which is on the first example sheet. One expects, and it is true, that $T(k) \to 1$ and $R(k) \to 0$ as $k \to \infty$: a sufficiently energetic wave barely notices the potential.

So far this is all hypothetical – we have described what a solution would look like if it existed. Let us check that it does.

The difficulty with existence questions for differential equations is that differentiation makes functions worse: it can destroy differentiability, or even continuity, so a differential operator may fling us out of whatever function space we started in. Integration, by contrast, makes functions better. So the standard move is to convert the differential equation into an integral equation, and work there.

Consider therefore the integral equation for $f = f(x,k)$,

$$f(x,k) = f_0(x,k) + \int_{-\infty}^{\infty} G(x-y,k)u(y)f(y,k)\,\mathrm{d}y,$$

where f_0 is any solution of $(\partial_x^2 + k^2)f_0 = 0$ and G is a Green's function for $\partial_x^2 + k^2$, so that

$$(\partial_x^2 + k^2)G(x,k) = \delta(x).$$

Any solution of the integral equation solves the eigenvalue problem, since

$$(\partial_x^2 + k^2)f = (\partial_x^2 + k^2)f_0 + \int_{-\infty}^{\infty} (\partial_x^2 + k^2)G(x-y,k)u(y)f(y,k)\,\mathrm{d}y$$

$$= 0 + \int_{-\infty}^{\infty} \delta(x-y)u(y)f(y,k)\,\mathrm{d}y$$
$$= u(x)f(x,k),$$

which rearranges to $Lf = k^2 f$.

To get the boundary condition we want, we choose $f_0 = e^{-ikx}$ and take the retarded Green's function

$$G(x,k) = \begin{cases} 0 & x < 0, \\ \frac{1}{k}\sin(kx) & x \geq 0. \end{cases}$$

Then the integrand vanishes unless $y \leq x$, so for x far to the left of the support of u the integral is zero and $f(x,k) = e^{-ikx}$ exactly, as required.

It remains to show that the integral equation has a solution. Write it abstractly as

$$(I - K)f = f_0, \qquad (Kf)(x) = \int_{-\infty}^{\infty} G(x-y,k)u(y)f(y,k)\,\mathrm{d}y,$$

so that formally

$$f = (I-K)^{-1}f_0 = \left(I + K + K^2 + \cdots\right) f_0.$$

This is a **Neumann series**, and one shows on the second example sheet that it converges (the point being that G is bounded and u has compact support, so K is a contraction after finitely many iterations). Given convergence, it visibly solves the equation:

$$(I-K)\left(f_0 + Kf_0 + K^2 f_0 + \cdots\right) = f_0 + Kf_0 + \cdots - \left(Kf_0 + K^2 f_0 + \cdots\right) = f_0.$$

We will use this series once more, for a completely different purpose, in Section 3.4; the key consequence to remember is that *the solution with* $f_0 = 0$ *is* $f = 0$.

§3.1.2 Bound States

Now take $\lambda = -\kappa^2 < 0$, and look for solutions of

$$L\psi_\kappa = -\kappa^2 \psi_\kappa$$

which are normalisable,

$$\|\psi_\kappa\|^2 = \int_{-\infty}^{\infty} \psi_\kappa(x)^2\,\mathrm{d}x = 1.$$

These are the **bound states**, and we take them real without loss of generality.

Again, since u has compact support, any solution obeys $\varphi_{xx} - \kappa^2\varphi = 0$ for large $|x|$, so is a combination of $e^{\pm\kappa x}$ out there. Fix φ_κ by

$$\varphi_\kappa(x) = e^{-\kappa x} \quad \text{as } x \to +\infty,$$

so that it decays to the right, and write

$$\varphi_\kappa(x) = \alpha(\kappa)e^{\kappa x} + \beta(\kappa)e^{-\kappa x} \quad \text{as } x \to -\infty.$$

For *any* κ we can solve the ODE and find such a φ_κ. What we cannot do for any κ is make it normalisable: the term $\beta(\kappa)e^{-\kappa x}$ blows up as $x \to -\infty$, so we need

$$\beta(\kappa) = 0.$$

This is a genuine constraint, and it is what quantises the spectrum. One can show that β has only finitely many zeros, which we label

$$\chi_1 > \chi_2 > \cdots > \chi_N > 0.$$

The corresponding normalised bound states are

$$\psi_n(x) = c_n \varphi_{\chi_n}(x),$$

with the constants c_n chosen so that $\|\psi_n\| = 1$. Equivalently, c_n is determined by the large-x behaviour

$$\psi_n(x) = c_n e^{-\chi_n x} \quad \text{as } x \to +\infty.$$

We call the c_n the **normalisation constants**.

§3.1.3 The Scattering Data

Assembling everything, the outcome of the forward problem is the following package.

Definition 3.2 (Scattering data)
The **scattering data** of the operator $L = -\partial_x^2 + u$ is

$$S = \left\{ \{\chi_n, c_n\}_{n=1}^N,\ R(k),\ T(k) \right\}.$$

Let us compute it in an example where everything can be done by hand.

Example 3.3 (A delta-function potential)
Take $u(x) = -2\alpha\delta(x)$ with $\alpha > 0$. (This is not of compact support, but it decays even faster, so all is well.)

Continuous spectrum. Since u vanishes away from the origin, we have exactly

$$\Phi(x,k) = \begin{cases} T(k)e^{-ikx} & x < 0, \\ e^{-ikx} + R(k)e^{ikx} & x > 0. \end{cases}$$

Continuity at $x = 0$ gives

$$T = 1 + R.$$

For the second condition, integrate $L\Phi = k^2\Phi$ over $(-\varepsilon, \varepsilon)$ and let $\varepsilon \to 0$. The $k^2\Phi$ term contributes nothing in the limit, and we obtain the jump condition

$$\Big[\Phi_x\Big]_{0^-}^{0^+} = -2\alpha\Phi(0),$$

that is

$$(-ik + ikR) - (-ikT) = -2\alpha T.$$

Substituting $T = 1 + R$ and solving the two equations gives

$$R(k) = \frac{i\alpha}{k - i\alpha}, \qquad T(k) = \frac{k}{k - i\alpha}.$$

As a check,

$$|R|^2 + |T|^2 = \frac{\alpha^2 + k^2}{k^2 + \alpha^2} = 1,$$

and as $k \to \infty$ we do indeed find $R \to 0$, $T \to 1$.

Bound states. Away from the origin $-\psi_{xx} + \chi^2 \psi = 0$, and decay at both ends forces

$$\psi(x) = c\, e^{-\chi |x|}.$$

The same jump condition now reads

$$-c\chi - c\chi = -2\alpha c \quad \Longrightarrow \quad \chi = \alpha.$$

So there is exactly one bound state, with $\chi_1 = \alpha$. Normalising,

$$1 = c_1^2 \int_{-\infty}^{\infty} e^{-2\alpha|x|}\, \mathrm{d}x = \frac{c_1^2}{\alpha} \quad \Longrightarrow \quad c_1 = \sqrt{\alpha}.$$

Hence the scattering data is

$$S = \left\{ \{\alpha, \sqrt{\alpha}\},\ \frac{i\alpha}{k - i\alpha},\ \frac{k}{k - i\alpha} \right\}.$$

The next example is the one that matters most, because it is a soliton.

Example 3.4 (The soliton potential)
Take $u(x) = -2\,\mathrm{sech}^2\, x$, which is the one-soliton profile with $\chi = 1$ at time $t = 0$.

Continuous spectrum. We claim that

$$f(x, k) = e^{ikx} \left(\tanh x - ik\right)$$

satisfies $Lf = k^2 f$. Writing $T = \tanh x$, so that $T' = \mathrm{sech}^2\, x = 1 - T^2$ and $T'' = -2TT'$,

$$\begin{aligned} f' &= e^{ikx} \left[ik(T - ik) + T' \right], \\ f'' &= e^{ikx} \left[-k^2(T - ik) + 2ikT' + T'' \right], \end{aligned}$$

and hence

$$\begin{aligned} -f'' - 2T'f &= e^{ikx} \left[k^2(T - ik) - 2ikT' - T'' - 2T'(T - ik) \right] \\ &= e^{ikx} \left[k^2(T - ik) - 2ikT' - T'' - 2TT' + 2ikT' \right] \\ &= e^{ikx} k^2 (T - ik), \end{aligned}$$

since $-T'' - 2TT' = 2TT' - 2TT' = 0$. So $Lf = k^2 f$ as claimed.

Replacing k by $-k$ gives another solution,

$$g(x, k) = e^{-ikx} \left(\tanh x + ik\right),$$

whose limits are

$$g \to e^{-ikx}(ik-1) \text{ as } x \to -\infty, \qquad g \to e^{-ikx}(ik+1) \text{ as } x \to +\infty.$$

So the normalised Jost solution is $\varphi = g/(ik-1)$, and comparing with the definitions,

$$a(k) = \frac{ik+1}{ik-1}, \qquad b(k) = 0.$$

Therefore

$$R(k) = 0, \qquad T(k) = \frac{ik-1}{ik+1}.$$

The reflection coefficient vanishes identically! A soliton is completely transparent: waves of every frequency pass through it without any reflection at all. (Note $|T| = 1$, consistent with $|R|^2 + |T|^2 = 1$.)

Bound states. We look for a normalisable solution of $-\psi_{xx} - 2\operatorname{sech}^2 x\,\psi = -\chi^2\psi$. Trying $\psi = \operatorname{sech} x$, and using $(\operatorname{sech} x)'' = \operatorname{sech} x - 2\operatorname{sech}^3 x$, we get

$$-\psi_{xx} - 2\operatorname{sech}^2 x\,\psi = -\operatorname{sech} x + 2\operatorname{sech}^3 x - 2\operatorname{sech}^3 x = -\operatorname{sech} x,$$

so $\psi = \operatorname{sech} x$ is a bound state with $\chi_1 = 1$. Normalising, $\int \operatorname{sech}^2 x \, \mathrm{d}x = 2$, so the normalised state is $\psi_1 = \frac{1}{\sqrt{2}} \operatorname{sech} x$. Since $\operatorname{sech} x \sim 2e^{-x}$ as $x \to +\infty$,

$$\psi_1(x) \sim \sqrt{2}\, e^{-x}, \qquad \text{so} \qquad c_1 = \sqrt{2}.$$

One can check there are no others. So the scattering data is simply

$$S = \left\{ \{1, \sqrt{2}\},\ R(k) = 0 \right\}.$$

Definition 3.5 (Reflectionless potential)

A potential u with $R(k) \equiv 0$ is called **reflectionless**.

We have just seen that the one-soliton potential is reflectionless. This is not a coincidence, and it is the key to everything: reflectionless potentials are exactly the potentials for which the inverse problem can be solved in closed form, and they will turn out to be exactly the N-soliton solutions.

§3.2 The Inverse Scattering Problem

Now we go backwards. Given scattering data S, can we reconstruct the potential u that produced it? The answer is yes, and remarkably the transmission coefficient is not even needed.

Theorem 3.6 (Gelfand–Levitan–Marchenko)

A potential $u = u(x)$ decaying rapidly as $|x| \to \infty$ is completely determined by the scattering data

$$S = \left\{ \{\chi_n, c_n\}_{n=1}^N,\ R(k) \right\}.$$

Specifically, define

$$F(x) = \sum_{n=1}^{N} c_n^2 e^{-\chi_n x} + \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} R(k)\, \mathrm{d}k,$$

and let $K(x, y)$ be the unique solution of the **GLM equation**

$$K(x, y) + F(x+y) + \int_x^\infty K(x, z) F(z+y)\, \mathrm{d}z = 0, \qquad y > x.$$

Then

$$u(x) = -2 \frac{\mathrm{d}}{\mathrm{d}x} K(x, x).$$

Proof. Beyond the scope of this course. □

The GLM equation looks forbidding but is not. For fixed x it is a linear integral equation in y, of the abstract form

$$\mathbf{K} + \mathbf{b} + \mathcal{A}\mathbf{K} = 0$$

for a known vector $\mathbf{b}$ and known linear operator $\mathcal{A}$. As with any linear equation, one can guess a solution and then verify it. In the reflectionless case, which is the one we shall care about, the integral equation degenerates into a finite-dimensional matrix equation and can be solved exactly.

Notice that the function F has a very transparent structure: the sum records the bound states, and the integral is the Fourier transform of the reflection coefficient. If there are no bound states and no reflection, $F = 0$, hence $K = 0$, hence $u = 0$ – reassuringly, the empty scattering data corresponds to the zero potential.

Example 3.7 (Reconstructing the soliton)

Let us check the theorem on the data we computed above,

$$S = \left\{ \{\chi_1, c_1\} = \{1, \sqrt{2}\},\ R = 0 \right\},$$

and see whether we recover $u = -2\,\mathrm{sech}^2\, x$. Here

$$F(x) = c_1^2 e^{-\chi_1 x} = 2e^{-x}.$$

Since F is a single exponential in x, we look for a solution of the same shape,

$$K(x, y) = K_1(x) e^{-y}.$$

Substituting into the GLM equation,

$$K_1(x) e^{-y} + 2e^{-x} e^{-y} + \int_x^\infty K_1(x) e^{-z} \cdot 2e^{-z} e^{-y}\, \mathrm{d}z = 0.$$

Every term carries a factor e^{-y}, which we cancel, and the remaining integral is

$$2K_1(x) \int_x^\infty e^{-2z}\, \mathrm{d}z = K_1(x) e^{-2x}.$$

So
$$K_1(x)\left(1+e^{-2x}\right) = -2e^{-x}, \qquad K_1(x) = \frac{-2e^{-x}}{1+e^{-2x}}.$$
Hence, writing $w = e^{-2x}$,
$$K(x,x) = K_1(x)e^{-x} = \frac{-2w}{1+w},$$
and since $\mathrm{d}w/\,\mathrm{d}x = -2w$,
$$\frac{\mathrm{d}}{\mathrm{d}x}K(x,x) = -2\cdot\frac{w'(1+w)-ww'}{(1+w)^2} = \frac{-2w'}{(1+w)^2} = \frac{4w}{(1+w)^2}.$$
Therefore
$$u(x) = -2\frac{\mathrm{d}}{\mathrm{d}x}K(x,x) = \frac{-8w}{(1+w)^2} = \frac{-8e^{-2x}}{(1+e^{-2x})^2} = \frac{-8}{(e^{x}+e^{-x})^2} = -2\operatorname{sech}^2 x,$$
exactly as it should be.

We can now go in both directions between potentials and scattering data. That, on its own, is a piece of nineteenth-century-flavoured mathematical physics. What makes it explosive is the next ingredient.

§3.3 Lax Pairs

We need to connect the time evolution of u to the time evolution of the scattering data. The mechanism is due to Peter Lax.

Recall from Section 1 that under a Hamiltonian flow, functions evolve according to
$$\frac{\mathrm{d}f}{\mathrm{d}t} = \{f, H\},$$
so H 'generates' the time evolution. In quantum mechanics the corresponding statement in the Heisenberg picture is
$$i\hbar\frac{\mathrm{d}L}{\mathrm{d}t} = [L, H],$$
for operators L. We want something of exactly this shape: an operator whose commutator with L generates the time evolution of L.

Definition 3.8 (Lax pair)

Let
$$L = a_m(x,t)\frac{\partial^m}{\partial x^m} + \cdots + a_1(x,t)\frac{\partial}{\partial x} + a_0(x,t)$$
be a time-dependent self-adjoint linear operator, and write
$$L_t = \dot{a}_m\frac{\partial^m}{\partial x^m} + \cdots + \dot{a}_0$$
for its derivative with respect to t (differentiating the coefficients only). If there is an operator A with
$$L_t = LA - AL = [L, A],$$

then (L, A) is called a **Lax pair**, and the equation $L_t = [L, A]$ is called **Lax's equation**.

The reason to care is the following.

Theorem 3.9 (Isospectral flow)
Let (L, A) be a Lax pair, with A antisymmetric. Then the discrete eigenvalues of L are independent of time. Moreover if $L\psi = \lambda\psi$ then

$$L\tilde{\psi} = \lambda\tilde{\psi}, \qquad \text{where } \tilde{\psi} = \psi_t + A\psi.$$

The word **isospectral** means precisely this: the operator changes with time, but its spectrum does not.

Proof. We assume that eigenvalues vary smoothly, so that an eigenvalue λ_0 with normalised eigenfunction ψ_0 at $t = 0$ extends to a family $\lambda(t)$, $\psi(x, t)$ with

$$L(t)\psi(x, t) = \lambda(t)\psi(x, t), \qquad \|\psi\| = 1.$$

Differentiate with respect to t:

$$L_t\psi + L\psi_t = \lambda_t\psi + \lambda\psi_t,$$

so, substituting Lax's equation and using $L\psi = \lambda\psi$,

$$\begin{aligned} \lambda_t\psi &= L_t\psi + L\psi_t - \lambda\psi_t \\ &= LA\psi - AL\psi + L\psi_t - \lambda\psi_t \\ &= LA\psi - \lambda A\psi + L\psi_t - \lambda\psi_t \\ &= (L - \lambda)\left(\psi_t + A\psi\right). \end{aligned}$$

Now take the inner product with ψ and use self-adjointness of $L - \lambda$:

$$\lambda_t = \langle\psi, \lambda_t\psi\rangle = \langle\psi, (L - \lambda)(\psi_t + A\psi)\rangle = \langle(L - \lambda)\psi, \psi_t + A\psi\rangle = 0,$$

since $(L - \lambda)\psi = 0$. So λ is constant, and the displayed equation reduces to $(L - \lambda)\tilde{\psi} = 0$. $\square$

So far this is abstract. The whole thing turns on being able to find, for a given equation, an operator A making Lax's equation equivalent to it. For KdV such an A exists, and here it is.

Proposition 3.10 (Lax pair for KdV)
Let

$$L = -\partial_x^2 + u(x, t), \qquad A = 4\partial_x^3 - 3\left(u\partial_x + \partial_x u\right).$$

Then A is antisymmetric, and

$$L_t = [L, A] \quad \Longleftrightarrow \quad u_t + u_{xxx} - 6uu_x = 0.$$

Proof. Throughout, $\partial = \partial_x$, and we treat everything as operators acting on a test function; in particular ∂u means the operator $\psi \mapsto (u\psi)_x$, so that

$$\partial u = u\partial + u_x.$$

Hence

$$A = 4\partial^3 - 3u\partial - 3\left(u\partial + u_x\right) = 4\partial^3 - 6u\partial - 3u_x.$$

For antisymmetry, recall $\partial^\dagger = -\partial$ and $u^\dagger = u$, so

$$A^\dagger = -4\partial^3 - 3\left(u\partial\right)^\dagger - 3u_x = -4\partial^3 + 3\partial u - 3u_x = -4\partial^3 + 3u\partial + 3u_x - 3u_x,$$

giving $A^\dagger = -4\partial^3 + 6u\partial + 3u_x = -A$ once we recombine (the middle term arises twice, once from each half of $u\partial + \partial u$).

Now for the commutator. We will repeatedly use the operator identities

$$\partial^2 u\partial = u\partial^3 + 2u_x\partial^2 + u_{xx}\partial, \qquad \partial^2 u_x = u_x\partial^2 + 2u_{xx}\partial + u_{xxx},$$

$$\partial^3 u = u\partial^3 + 3u_x\partial^2 + 3u_{xx}\partial + u_{xxx},$$

each of which is just the Leibniz rule. First,

$$\begin{aligned} LA &= \left(-\partial^2 + u\right)\left(4\partial^3 - 6u\partial - 3u_x\right) \\ &= -4\partial^5 + 6\partial^2 u\partial + 3\partial^2 u_x + 4u\partial^3 - 6u^2\partial - 3uu_x \\ &= -4\partial^5 + 6\left(u\partial^3 + 2u_x\partial^2 + u_{xx}\partial\right) + 3\left(u_x\partial^2 + 2u_{xx}\partial + u_{xxx}\right) \\ &\quad + 4u\partial^3 - 6u^2\partial - 3uu_x \\ &= -4\partial^5 + 10u\partial^3 + 15u_x\partial^2 + 12u_{xx}\partial - 6u^2\partial + 3u_{xxx} - 3uu_x. \end{aligned}$$

Similarly,

$$\begin{aligned} AL &= \left(4\partial^3 - 6u\partial - 3u_x\right)\left(-\partial^2 + u\right) \\ &= -4\partial^5 + 4\partial^3 u + 6u\partial^3 - 6u\partial u + 3u_x\partial^2 - 3uu_x \\ &= -4\partial^5 + 4\left(u\partial^3 + 3u_x\partial^2 + 3u_{xx}\partial + u_{xxx}\right) + 6u\partial^3 \\ &\quad - 6\left(u^2\partial + uu_x\right) + 3u_x\partial^2 - 3uu_x \\ &= -4\partial^5 + 10u\partial^3 + 15u_x\partial^2 + 12u_{xx}\partial - 6u^2\partial + 4u_{xxx} - 9uu_x. \end{aligned}$$

Every differential term cancels in the difference, and we are left with the multiplication operator

$$[L, A] = LA - AL = (3u_{xxx} - 3uu_x) - (4u_{xxx} - 9uu_x) = -u_{xxx} + 6uu_x.$$

On the other hand $L_t = u_t$, since only the coefficient u depends on t. So Lax's equation $L_t = [L, A]$ says exactly

$$u_t = -u_{xxx} + 6uu_x,$$

which is KdV. □

This is the miracle on which everything rests. It deserves a moment's contemplation: a fifth-order operator equation collapses, after everything cancels, to a scalar equation – and that scalar equation happens to be KdV. Combined with the isospectral flow theorem, we already know something remarkable.

Corollary 3.11

If $u(x,t)$ evolves according to KdV, then the discrete eigenvalues $-\chi_n^2$ of the operator $L = -\partial_x^2 + u(x,t)$ are constant in time. In particular the number N of bound states never changes.

Since the χ_n are the soliton amplitudes and speeds, this already tells us that solitons cannot be created or destroyed by the evolution.

§3.4 Evolution of the Scattering Data

We now let $u = u(x,t)$ evolve by KdV and work out how the rest of the scattering data moves. Since u has compact support, note first that

$$A = 4\partial_x^3 \quad \text{as } |x| \to \infty.$$

§3.4.1 Continuous Spectrum

For each fixed t construct, as in Section 3.1, the solution φ of $L\varphi = k^2\varphi$ with

$$\varphi(x,t) = \begin{cases} e^{-ikx} & x \to -\infty, \\ a(k,t)e^{-ikx} + b(k,t)e^{ikx} & x \to +\infty. \end{cases}$$

Here k is fixed: solutions exist for every real k regardless of u, so there is no question of k moving.

Exactly as in the proof of the isospectral flow theorem (but now with $\lambda = k^2$ constant by fiat rather than by conclusion), differentiating $L\varphi = k^2\varphi$ gives

$$0 = (L - k^2)\left(\varphi_t + A\varphi\right),$$

so $\tilde{\varphi} = \varphi_t + A\varphi$ is another solution of $L\tilde{\varphi} = k^2\tilde{\varphi}$. We now compute its asymptotics, using $A = 4\partial_x^3$ out there:

$$\tilde{\varphi}(x,t) = \begin{cases} 4ik^3 e^{-ikx} & x \to -\infty, \\ \left(a_t + 4ik^3 a\right)e^{-ikx} + \left(b_t - 4ik^3 b\right)e^{ikx} & x \to +\infty, \end{cases}$$

since $\partial_x^3 e^{\mp ikx} = \pm ik^3 e^{\mp ikx}$.

Here is the clever step. Consider

$$\theta = 4ik^3\varphi - \tilde{\varphi}.$$

By linearity $L\theta = k^2\theta$, and by construction $\theta \to 0$ as $x \to -\infty$. But we showed in Section 3.1 that the solution of $Lf = k^2 f$ with prescribed behaviour f_0 at $-\infty$ is $f = (I + K + K^2 + \cdots)f_0$; with $f_0 = 0$ this gives $f = 0$. Hence

$$\theta \equiv 0, \qquad \text{that is} \qquad \tilde{\varphi} = 4ik^3\varphi.$$

Comparing coefficients of e^{-ikx} and e^{ikx} as $x \to +\infty$,

$$a_t + 4ik^3 a = 4ik^3 a,$$

$$b_t - 4ik^3 b = 4ik^3 b,$$

that is $a_t = 0$ and $b_t = 8ik^3 b$. These we can solve:

$$a(k,t) = a(k,0), \qquad b(k,t) = b(k,0)e^{8ik^3 t},$$

and therefore

$$\boxed{R(k,t) = R(k,0)e^{8ik^3 t}, \qquad T(k,t) = T(k,0).}$$

Stop and appreciate this. We assumed that u evolves by a thoroughly nasty nonlinear PDE, and concluded that its reflection coefficient simply rotates in the complex plane at a constant rate, and that its transmission coefficient does not move at all.

§3.4.2 Discrete Spectrum

By the isospectral flow theorem the χ_n are constant. For each t we have normalised bound states $\psi_n(x,t)$ with

$$L\psi_n = -\chi_n^2 \psi_n, \qquad \|\psi_n\| = 1, \qquad \psi_n(x,t) = c_n(t)e^{-\chi_n x} \quad \text{as } x \to +\infty.$$

By the isospectral flow theorem again,

$$\tilde{\psi}_n = \partial_t \psi_n + A\psi_n$$

also satisfies $L\tilde{\psi}_n = -\chi_n^2 \tilde{\psi}_n$. Since the eigenvalue $-\chi_n^2$ is simple (a standard Wronskian argument shows bound states of a one-dimensional Schrödinger operator are non-degenerate), $\tilde{\psi}_n$ must be proportional to ψ_n. But

$$\begin{aligned}
\langle \psi_n, \tilde{\psi}_n \rangle &= \langle \psi_n, \partial_t \psi_n \rangle + \langle \psi_n, A\psi_n \rangle \\
&= \frac{1}{2}\frac{\partial}{\partial t}\langle \psi_n, \psi_n \rangle + \langle \psi_n, A\psi_n \rangle \\
&= 0,
\end{aligned}$$

because $\|\psi_n\|$ is constant and A is antisymmetric (so $\langle \psi, A\psi \rangle = -\langle A\psi, \psi \rangle = -\langle \psi, A\psi \rangle$). A multiple of ψ_n orthogonal to ψ_n is zero, so

$$\tilde{\psi}_n \equiv 0.$$

Now look at the large-x behaviour. There $A = 4\partial_x^3$ and $\psi_n = c_n(t)e^{-\chi_n x}$, so

$$\tilde{\psi}_n(x,t) = \left(\dot{c}_n - 4\chi_n^3 c_n\right) e^{-\chi_n x} \quad \text{as } x \to +\infty.$$

Since $\tilde{\psi}_n \equiv 0$ we must have $\dot{c}_n = 4\chi_n^3 c_n$, and hence

$$\boxed{c_n(t) = c_n(0)e^{4\chi_n^3 t}.}$$

§3.4.3 The Transform

We can now assemble the machine. Suppose u evolves by KdV with initial data $u_0(x) = u(x,0)$.

$$\begin{tikzcd}[column sep=huge, row sep=large]
u_0(x) \arrow[r, "\text{scattering problem}"] \arrow[d, dashed, "\text{KdV equation}"'] & S(0) = \big\{\{\chi_n, c_n(0)\}_{n=1}^N,\ R(k,0)\big\} \arrow[d, "\text{evolve the scattering data}"] \\
u(x,t) & S(t) = \left\{\{\chi_n, c_n(0)e^{4\chi_n^3 t}\}_{n=1}^N,\ R(k,0)e^{8ik^3 t}\right\} \arrow[l, "\text{solve the GLM equation}"]
\end{tikzcd}$$

That is: compute the scattering data of u_0; evolve it by the explicit formulae above; feed the result into the GLM equation; and read off $u(x,t)$. Compare this with the Fourier transform diagram at the start of this section – it is the same three steps, with a more sophisticated transform. All the nonlinearity of KdV has been absorbed into the (nonlinear, but classical) map $u \mapsto S$.

The one thing that makes it work is the equivalence

$$u_t + u_{xxx} - 6uu_x = 0 \quad\Longleftrightarrow\quad L_t = [L, A],$$

so the whole method hinges on finding a Lax pair.

Remark. This is not special to KdV. Choosing different operators A in the Lax equation with the same L produces an infinite hierarchy of equations – the **KdV hierarchy** – all of which are solvable by the same transform and all of which share the same conserved quantities. Choosing a different L altogether produces different integrable equations; the choice

$$L = i\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\partial_x + \begin{pmatrix} 0 & q \\ \bar{q} & 0 \end{pmatrix}$$

leads, by the same route, to the nonlinear Schrödinger equation, which we meet in Section 5.

§3.5 Reflectionless Potentials and N-Solitons

Now we cash in. The GLM equation is hard in general, but there is one case in which it can be solved completely, and it is exactly the case that produces solitons.

Recall that a potential is reflectionless if $R(k,0) = 0$. Since

$$R(k,t) = R(k,0)e^{8ik^3 t},$$

a potential which starts reflectionless stays reflectionless forever. So this is a class of initial data preserved by the KdV flow, and within it the machinery becomes elementary.

Suppose then that $R \equiv 0$, so that

$$F(x) = \sum_{n=1}^{N} c_n^2 e^{-\chi_n x}.$$

(We suppress the time dependence for the moment, restoring it at the very end.) Since F is a finite sum of exponentials, we look for K of the same shape,

$$K(x,y) = \sum_{m=1}^{N} K_m(x) e^{-\chi_m y};$$

one shows on the second example sheet that K *must* have this form. Substituting into the GLM equation,

$$\sum_{m=1}^{N} K_m(x)e^{-\chi_m y} + \sum_{n=1}^{N} c_n^2 e^{-\chi_n x}e^{-\chi_n y} + \sum_{n,m} c_n^2 K_m(x) \int_x^\infty e^{-\chi_m z}e^{-\chi_n (z+y)}\, \mathrm{d}z = 0.$$

The integral evaluates to

$$\int_x^\infty e^{-(\chi_n+\chi_m)z}\, \mathrm{d}z = \frac{e^{-(\chi_n+\chi_m)x}}{\chi_n+\chi_m},$$

so, collecting everything over the common index n,

$$\sum_{n=1}^{N} \left[c_n^2 e^{-\chi_n x} + K_n(x) + \sum_{m=1}^{N} \frac{c_n^2 K_m(x)}{\chi_n+\chi_m} e^{-(\chi_n+\chi_m)x} \right] e^{-\chi_n y} = 0. \tag{$*$}$$

The functions $e^{-\chi_n y}$, $n = 1, \dots, N$, are linearly independent, so each bracket must vanish separately. Thus for each n,

$$K_n(x) + \sum_{m=1}^{N} \frac{c_n^2 e^{-(\chi_n+\chi_m)x}}{\chi_n+\chi_m} K_m(x) = -c_n^2 e^{-\chi_n x}.$$

For each fixed x this is a linear system! Setting

$$\begin{aligned} \mathbf{c} &= \left(c_1^2 e^{-\chi_1 x}, \dots, c_N^2 e^{-\chi_N x}\right)^T, \\ \mathbf{K} &= \left(K_1(x), \dots, K_N(x)\right)^T, \\ A_{nm} &= \delta_{nm} + \frac{c_n^2 e^{-(\chi_n+\chi_m)x}}{\chi_n+\chi_m}, \end{aligned}$$

the system reads

$$A\mathbf{K} = -\mathbf{c}.$$

So we have reduced an integral equation to inverting an $N \times N$ matrix. Even better, we do not need $\mathbf{K}$ itself, only the diagonal value

$$K(x,x) = \sum_{m=1}^{N} K_m(x)e^{-\chi_m x} = \sum_{m,n} \left(A^{-1}\right)_{mn} (-\mathbf{c})_n\, e^{-\chi_m x}.$$

Now observe that

$$A'_{nm}(x) = \frac{\mathrm{d}}{\mathrm{d}x} A_{nm}(x) = -c_n^2 e^{-\chi_n x} e^{-\chi_m x} = (-\mathbf{c})_n\, e^{-\chi_m x},$$

which is exactly the combination appearing above. Hence

$$K(x,x) = \sum_{m,n} \left(A^{-1}\right)_{mn} A'_{nm} = \operatorname{tr}\left(A^{-1}A'\right).$$

Finally we use the identity (an exercise on the second example sheet, proved by expanding the determinant)

$$\operatorname{tr}\left(A^{-1}A'\right) = \frac{1}{\det A} \frac{\mathrm{d}}{\mathrm{d}x} (\det A) = \frac{\mathrm{d}}{\mathrm{d}x} \log(\det A),$$

which gives

$$u(x) = -2\frac{\mathrm{d}}{\mathrm{d}x} K(x,x) = -2\frac{\mathrm{d}^2}{\mathrm{d}x^2} \log(\det A).$$

Restoring the time dependence via $c_n(t)^2 = c_n(0)^2 e^{8\chi_n^3 t}$, we have proved the following.

Theorem 3.12 (N-soliton solution)
Let $\chi_1 > \cdots > \chi_N > 0$ and $c_1(0), \ldots, c_N(0)$ be given. Then

$$u(x,t) = -2\frac{\partial^2}{\partial x^2}\log\left(\det A(x,t)\right),$$

where A is the $N \times N$ matrix

$$A_{nm}(x,t) = \delta_{nm} + \frac{c_n(0)^2 e^{8\chi_n^3 t} e^{-(\chi_n+\chi_m)x}}{\chi_n + \chi_m},$$

is a solution of the KdV equation. It is the N-soliton solution, and describes N solitary waves of amplitudes $2\chi_n^2$ and speeds $4\chi_n^2$.

We have obtained a closed-form solution to a nonlinear PDE, involving nothing worse than a determinant. Let us unpack it in the two simplest cases.

Example 3.13 ($N = 1$)
With a single bound state, A is a 1×1 matrix,

$$\det A = 1 + \frac{c_1(0)^2 e^{8\chi_1^3 t} e^{-2\chi_1 x}}{2\chi_1} = 1 + e^{-2\chi_1(x - 4\chi_1^2 t - \delta)},$$

where the constant δ absorbs $c_1(0)$ via $e^{2\chi_1\delta} = c_1(0)^2/2\chi_1$. Writing $\xi = x - 4\chi_1^2 t - \delta$ and $w = e^{-2\chi_1\xi}$,

$$\log\det A = \log(1+w),$$

and differentiating twice (using $w' = -2\chi_1 w$),

$$\frac{\partial}{\partial x}\log(1+w) = \frac{-2\chi_1 w}{1+w}, \qquad \frac{\partial^2}{\partial x^2}\log(1+w) = \frac{4\chi_1^2 w}{(1+w)^2}.$$

Hence

$$u = -\frac{8\chi_1^2 w}{(1+w)^2} = -\frac{8\chi_1^2}{\left(w^{1/2} + w^{-1/2}\right)^2} = -\frac{8\chi_1^2}{4\cosh^2(\chi_1\xi)} = -2\chi_1^2\operatorname{sech}^2\left(\chi_1(x - 4\chi_1^2 t - \delta)\right),$$

which is precisely the one-soliton solution of Section 2.3. So a single bound state means a single soliton, with χ_1 determining both its height and its speed, and $c_1(0)$ determining only where it is.

Example 3.14 ($N = 2$)
With two bound states, write

$$E_n = E_n(x,t) = \frac{c_n(0)^2 e^{8\chi_n^3 t} e^{-2\chi_n x}}{2\chi_n},$$

so that $A_{nn} = 1 + E_n$ and

$$A_{12}A_{21} = \frac{c_1^2 c_2^2 e^{-2(\chi_1+\chi_2)x}}{(\chi_1+\chi_2)^2} = \frac{4\chi_1\chi_2}{(\chi_1+\chi_2)^2} E_1 E_2.$$

Therefore

$$\det A = (1+E_1)(1+E_2) - \frac{4\chi_1\chi_2}{(\chi_1+\chi_2)^2} E_1 E_2 = 1 + E_1 + E_2 + \left(\frac{\chi_2-\chi_1}{\chi_2+\chi_1}\right)^2 E_1 E_2,$$

using $1 - \frac{4\chi_1\chi_2}{(\chi_1+\chi_2)^2} = \frac{(\chi_2-\chi_1)^2}{(\chi_2+\chi_1)^2}$. Substituting into the theorem gives the two-soliton solution plotted in Section 2.4.

The interesting thing is what happens at large $|t|$, and this we can extract without any further computation. Write

$$\delta = \frac{\chi_2-\chi_1}{\chi_2+\chi_1} \in (0,1),$$

and follow the taller soliton by fixing $\xi = x - 4\chi_2^2 t$. In this frame

$$E_2 \propto e^{8\chi_2^3 t - 2\chi_2(4\chi_2^2 t + \xi)} = e^{-2\chi_2\xi},$$

which is constant in t, while

$$E_1 \propto e^{8\chi_1^3 t - 2\chi_1(4\chi_2^2 t+\xi)} = e^{8\chi_1(\chi_1^2-\chi_2^2)t} e^{-2\chi_1\xi} \longrightarrow \begin{cases} 0 & t \to +\infty, \\ \infty & t \to -\infty, \end{cases}$$

since $\chi_1 < \chi_2$. So:

- As $t \to +\infty$ we have $\det A \to 1 + E_2$, and by the $N = 1$ computation u tends to a soliton of parameter χ_2 sitting at $\xi = \delta_2$ for some fixed δ_2.
- As $t \to -\infty$ the term E_1 dominates, and

  $$\det A \approx E_1\left(1 + \delta^2 E_2\right).$$

  Now $\log E_1$ is a linear function of x, so it is killed by ∂_x^2 and contributes nothing. We are left with $u \to -2\partial_x^2 \log(1+\delta^2 E_2)$, which is again a soliton of parameter χ_2 – but with E_2 replaced by $\delta^2 E_2$.

Since $E_2 \propto e^{-2\chi_2\xi}$, multiplying E_2 by δ^2 is the same as translating ξ by

$$\Delta \quad \text{where} \quad e^{2\chi_2\Delta} = \delta^2, \quad \text{i.e.} \quad \Delta = \frac{1}{\chi_2}\log\delta = -\frac{1}{\chi_2}\log\frac{\chi_2+\chi_1}{\chi_2-\chi_1} < 0.$$

So long before the collision the tall soliton is at $\xi = \delta_2 + \Delta$, and long afterwards it is at $\xi = \delta_2$: it has been pushed *forward* by

$$|\Delta| = \frac{1}{\chi_2}\log\frac{\chi_2+\chi_1}{\chi_2-\chi_1}.$$

Repeating the argument in the frame of the shorter soliton (where now E_2 is the one that blows up) gives a shift backwards by $\frac{1}{\chi_1}\log\frac{\chi_2+\chi_1}{\chi_2-\chi_1}$. These are the phase shifts we quoted in Section 2.4.

Two remarks are in order. First, the number of solitons is the number of bound states, so we can read off the eventual fate of *any* initial profile from a linear eigenvalue problem: count the bound states of $-\partial_x^2 + u_0$, and that is how many solitons will emerge. Everything else – the reflection coefficient – disperses away, and one can show that it contributes a decaying oscillatory tail travelling in the opposite direction. So the general picture is:

any decaying initial profile resolves into N solitons plus radiation.

Second, the total absence of any interaction term in the final answer – the shifts Δ depend only on χ_1 and χ_2, not on when or where the collision happened – is precisely the sense in which solitons behave like particles.

§3.6 Infinitely Many Conservation Laws

We promised at the start of Section 2 that an integrable PDE should have infinitely many conserved quantities, and in Section 2.5 we found three of them by inspection. Now we can produce them all at once.

The idea is that we have already found something conserved: the quantity $a(k,t) = a(k,0)$ is independent of time, for every k. If we can expand $a(k)$ in powers of $1/k$, then every coefficient in that expansion must be conserved as well, and we will have infinitely many first integrals in a single stroke.

Recall the setup: $\varphi = \varphi(x,k,t)$ solves $L\varphi = k^2\varphi$ with

$$\varphi = \begin{cases} e^{-ikx} & x \to -\infty, \\ a(k,t)e^{-ikx} + b(k,t)e^{ikx} & x \to +\infty. \end{cases}$$

Multiplying by e^{ikx},

$$e^{ikx}\varphi(x,k,t) = a(k) + b(k,t)e^{2ikx} \quad \text{as } x \to +\infty.$$

To isolate a we average out the oscillating term. Over the window $[R, 2R]$,

$$\frac{1}{R}\int_R^{2R} a(k)\,\mathrm{d}x = a(k), \qquad \frac{1}{R}\int_R^{2R} b(k,t)e^{2ikx}\,\mathrm{d}x = O\left(\frac{1}{R}\right),$$

so

$$a(k) = \lim_{R\to\infty} \frac{1}{R}\int_R^{2R} e^{ikx}\varphi(x,k,t)\,\mathrm{d}x = \lim_{R\to\infty}\int_1^2 e^{ikRx}\varphi(Rx,k,t)\,\mathrm{d}x,$$

substituting $x \mapsto Rx$.

Now we need a good representation of φ. Since $\varphi = e^{-ikx}$ as $x \to -\infty$, it is natural to write

$$\varphi(x,k,t) = \exp\left(-ikx + \int_{-\infty}^{x} S(y,k,t)\,\mathrm{d}y\right)$$

for some function S. Feeding this into the formula above and passing to the limit,

$$a(k) = \lim_{R\to\infty}\int_1^2 \exp\left(\int_{-\infty}^{Rx} S(y,k,t)\,\mathrm{d}y\right)\mathrm{d}x = \exp\left(\int_{-\infty}^{\infty} S(y,k,t)\,\mathrm{d}y\right). \qquad (\dagger)$$

The left-hand side has no t in it; the right-hand side appears to. That is where our conserved quantities will come from.

It remains to find S. Substituting the exponential ansatz into $L\varphi = k^2\varphi$, and using

$$\varphi_x = (S - ik)\varphi, \qquad \varphi_{xx} = S_x\varphi + (S - ik)^2\varphi,$$

we find, after dividing by φ,

$$S_x - 2ikS + S^2 = -u.$$

The function φ has disappeared entirely, and we are left with a **Riccati equation** for S. (This is the Riccati equation of the Miura transform in disguise, which is no coincidence.)

We solve it as an asymptotic series in $1/k$:

$$S(x, k, t) = \sum_{n=1}^{\infty} \frac{S_n(x, t)}{(2ik)^n}.$$

Substituting and comparing coefficients of k^{-n} gives the recursion

$$S_1 = -u, \qquad S_{n+1} = \frac{\mathrm{d}S_n}{\mathrm{d}x} + \sum_{m=1}^{n-1} S_m S_{n-m}.$$

This is entirely mechanical, and the first few terms are

$$S_1 = -u, \qquad S_2 = -u_x, \qquad S_3 = -u_{xx} + u^2, \qquad S_4 = -u_{xxx} + 4uu_x, \ \ldots$$

Now put this back into (†):

$$\log a(k) = \int_{-\infty}^{\infty} S(x, k, t)\,\mathrm{d}x = \sum_{n=1}^{\infty} \frac{1}{(2ik)^n} \int_{-\infty}^{\infty} S_n(x, t)\,\mathrm{d}x.$$

The left-hand side is independent of t, and this holds for all k, so every coefficient must be independent of t separately. We conclude:

Theorem 3.15

If u evolves by KdV then, for every $n \geq 1$,

$$\int_{-\infty}^{\infty} S_n(x, t)\,\mathrm{d}x$$

is a first integral, where the S_n are given by the recursion above.

Let us look at the first few.

(i) $n = 1$ gives

$$\int_{-\infty}^{\infty} u\,\mathrm{d}x,$$

conservation of mass – our I_1.

(ii) $n = 2$ gives $\int u_x\,\mathrm{d}x$, which vanishes identically because u decays. This is a *trivial* conservation law: true, but empty. In fact one can show that S_n is a total derivative for every even n, so the even members of the family give us nothing.

(iii) $n = 3$ gives

$$\int_{-\infty}^{\infty} \left(-u_{xx} + u^2\right)\,\mathrm{d}x = \int_{-\infty}^{\infty} u^2\,\mathrm{d}x,$$

conservation of momentum – our I_2.

(iv) $n = 5$ gives $S_5 = -u_{xxxx} + 5u_x^2 + 6uu_{xx} - 2u^3$. Integrating, the first term vanishes and $\int 6uu_{xx}\, \mathrm{d}x = -\int 6u_x^2\, \mathrm{d}x$ by parts, leaving

$$\int_{-\infty}^{\infty} S_5\, \mathrm{d}x = -\int_{-\infty}^{\infty} \left(u_x^2 + 2u^3\right)\, \mathrm{d}x = -2I_3,$$

so we recover the energy.

So half of the family is trivial. But half of infinity is still infinity, and we have produced infinitely many genuine first integrals of KdV, exactly as the finite-dimensional theory led us to expect.

Remark. It is worth noticing where the conserved quantities came from: they are the coefficients in an expansion of the transmission data. The scattering data $\{\chi_n\}$ and $R(k)$ are, in a real sense, the action variables of KdV; the phases of c_n and R are the angles, moving at constant rates $4\chi_n^3$ and $8k^3$. In the next section we make this analogy considerably more precise.

§4 The Hamiltonian Structure of Integrable PDEs

The integrable ODEs of Section 1 were not arbitrary ODEs – they were Hamiltonian systems, and their integrability was expressed in terms of Poisson-commuting first integrals. If we really want to think of PDEs as infinite-dimensional dynamical systems, we should ask whether they too are Hamiltonian, and whether the conserved quantities we found are in involution. They are, and putting the theory in this language explains where the infinite family of conservation laws comes from.

§4.1 Infinite-Dimensional Hamiltonian Systems

Recall the finite-dimensional set-up: a phase space $\mathbb{R}^{2n}$, a non-degenerate antisymmetric matrix J, a Hamiltonian $H : \mathbb{R}^{2n} \to \mathbb{R}$, and the equation of motion

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = J\frac{\partial H}{\partial \mathbf{x}}.$$

We now promote each ingredient in turn. The rule of thumb is that the discrete index i becomes the continuous variable x, sums become integrals, and partial derivatives become functional derivatives.

First, the inner product. In finite dimensions we had $\mathbf{x} \cdot \mathbf{y} = \sum_i x_i y_i$; here we replace the sum by an integral.

Notation. For functions u, v on $\mathbb{R}$ we write

$$\langle u, v\rangle = \int_{\mathbb{R}} u(x)v(x)\, \mathrm{d}x.$$

If u and v also depend on t, then so does the inner product.

Secondly, functions on phase space. In finite dimensions the Hamiltonian was a function $H(\mathbf{x})$; here the analogue is not a function of u at each point, but a *functional.*

Definition 4.1 (Functional)
A **functional** is a real-valued map on a space of functions of the form

$$F[u] = \int_{\mathbb{R}} f(x, u, u_x, u_{xx}, \dots)\, \mathrm{d}x.$$

The function f is called the **density**.

Thirdly, differentiation with respect to phase-space coordinates. We met this in IB Variational Principles.

Definition 4.2 (Functional derivative)
The **functional derivative** (or Euler–Lagrange derivative) of F at u is the unique function δF with

$$\langle \delta F, \eta \rangle = \lim_{\varepsilon \to 0} \frac{F[u + \varepsilon\eta] - F[u]}{\varepsilon}$$

for all smooth η of compact support; equivalently

$$F[u + \varepsilon\eta] = F[u] + \varepsilon\langle \delta F, \eta \rangle + o(\varepsilon).$$

Note that δF is itself a function of x, depending on u. We often write $\delta F / \delta u$ for it.

Example 4.3
Let $F[u] = \frac{1}{2}\int u_x^2\, \mathrm{d}x$. Then

$$\begin{aligned} F[u + \varepsilon\eta] &= \frac{1}{2}\int (u_x + \varepsilon\eta_x)^2\, \mathrm{d}x \\ &= \frac{1}{2}\int u_x^2\, \mathrm{d}x + \varepsilon\int u_x\eta_x\, \mathrm{d}x + o(\varepsilon) \\ &= F[u] + \varepsilon\langle u_x, \eta_x \rangle + o(\varepsilon). \end{aligned}$$

This is not yet of the required form, since we want η rather than η_x in the second slot. When in doubt, integrate by parts – and since η has compact support there are no boundary terms:

$$F[u + \varepsilon\eta] = F[u] + \varepsilon\langle -u_{xx}, \eta \rangle + o(\varepsilon).$$

So $\delta F = -u_{xx}$.

The general formula, from IB Variational Principles, is

$$\delta F = \frac{\partial f}{\partial u} - \mathrm{D}_x\left(\frac{\partial f}{\partial u_x}\right) + \mathrm{D}_x^2\left(\frac{\partial f}{\partial u_{xx}}\right) - \cdots,$$

where D_x is the *total* derivative, not the partial one.

Definition 4.4 (Total derivative)

For a function $f(x, u, u_x, \dots)$ and a given $u = u(x)$, the **total derivative** is

$$\mathrm{D}_x f = \frac{\partial f}{\partial x} + u_x \frac{\partial f}{\partial u} + u_{xx} \frac{\partial f}{\partial u_x} + \cdots .$$

Example 4.5

For $f = xu$ we have $\partial f/\partial x = u$ but $\mathrm{D}_x f = u + xu_x$. The partial derivative treats u as an independent coordinate; the total derivative remembers that u is a function of x.

Finally we need an analogue of J. It is helpful first to notice that in finite dimensions we can think of J in two ways: as an antisymmetric bilinear form, or – since we also have a dot product – as an antisymmetric linear *map*, applied via $\mathbf{v} \cdot J\mathbf{w}$. It is the second picture that generalises: we look for an antisymmetric linear operator $\mathcal{J}$ acting on functions, and define

$$\{F, G\} = \langle \delta F, \mathcal{J} \delta G \rangle = \int \delta F(x)\, \mathcal{J} \delta G(x)\, \mathrm{d}x.$$

Bilinearity and antisymmetry are then automatic. The Jacobi identity, however, is *not*: it constrains the choice of $\mathcal{J}$.

Definition 4.6 (Hamiltonian operator)

A **Hamiltonian operator** is a linear, antisymmetric operator $\mathcal{J}$ on functions such that the induced bracket

$$\{F, G\} = \langle \delta F, \mathcal{J} \delta G \rangle$$

satisfies the Jacobi identity.

The simplest antisymmetric operator one can write down is $\mathcal{J} = \partial_x$ (antisymmetric because $\langle u_x, v \rangle = -\langle u, v_x \rangle$ after integrating by parts). One can check – the proof is straightforward but tedious – that it is Hamiltonian.

Definition 4.7 (Hamiltonian form)

An evolution equation for $u(x,t)$ is in **Hamiltonian form** if it can be written

$$u_t = \mathcal{J} \frac{\delta H}{\delta u}$$

for some functional $H[u]$ and some Hamiltonian operator $\mathcal{J}$.

Everything from Section 1 now goes through verbatim.

Proposition 4.8

If $u_t = \mathcal{J} \delta H$ and $I = I[u]$ is a functional, then

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \{I, H\}.$$

In particular I is a first integral if and only if $\{I, H\} = 0$.

Proof. Directly from the definition of the functional derivative,

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \lim_{\varepsilon \to 0} \frac{I[u+\varepsilon u_t] - I[u]}{\varepsilon} = \langle \delta I, u_t \rangle = \langle \delta I, \mathcal{J}\delta H \rangle = \{I, H\}. \qquad \square$$

In summary, the dictionary is:

$2n$-dimensional phase space	*infinite-dimensional phase space*
$x_i(t)$, $i = 1, \ldots, 2n$	$u(x,t)$, $x \in \mathbb{R}$
$\mathbf{x} \cdot \mathbf{y} = \sum_i x_i y_i$	$\langle u, v \rangle = \int uv \, \mathrm{d}x$
$\mathrm{d}/\,\mathrm{d}t$	$\partial/\partial t$
$\partial/\partial \mathbf{x}$	$\delta/\delta u$
antisymmetric matrix J	antisymmetric operator $\mathcal{J}$
functions $f(\mathbf{x})$	functionals $F[u]$

§4.2 Bi-Hamiltonian Systems

So far this is only a change of language. The interesting phenomenon is that the *same* equation may be Hamiltonian in more than one way.

Definition 4.9 (Bi-Hamiltonian)

An evolution equation is **bi-Hamiltonian** if it can be written in Hamiltonian form for two different Hamiltonian operators,

$$u_t = \mathcal{J}_0 \delta H_0 = \mathcal{J}_1 \delta H_1.$$

Example 4.10 (KdV is bi-Hamiltonian)

Take

$$\mathcal{J}_1 = \partial_x, \qquad H_1[u] = \int \left(\tfrac{1}{2}u_x^2 + u^3\right) \, \mathrm{d}x.$$

Then the density is $f = \frac{1}{2}u_x^2 + u^3$, and

$$\delta H_1 = \frac{\partial f}{\partial u} - \mathrm{D}_x \left(\frac{\partial f}{\partial u_x}\right) = 3u^2 - u_{xx},$$

so

$$\mathcal{J}_1 \delta H_1 = \partial_x \left(3u^2 - u_{xx}\right) = 6uu_x - u_{xxx} = u_t,$$

which is KdV. So H_1 – which is precisely the first integral I_3 from Section 2.5 – is the Hamiltonian for KdV.

But KdV is also Hamiltonian with respect to a completely different structure. Take

$$\mathcal{J}_0 = -\partial_x^3 + 4u\partial_x + 2u_x, \qquad H_0[u] = \int \tfrac{1}{2}u^2 \, \mathrm{d}x.$$

Now $\delta H_0 = u$, and

$$\mathcal{J}_0 \delta H_0 = -u_{xxx} + 4uu_x + 2u_x u = -u_{xxx} + 6uu_x = u_t,$$

which is KdV again. Here $H_0 = \frac{1}{2}I_2$ is the momentum. One checks that $\mathcal{J}_0$ is antisymmetric (the operators ∂_x^3 and $u\partial_x + \partial_x u$ both are) and that it satisfies the Jacobi identity, which is less obvious.

Why does this matter? Because two compatible Hamiltonian structures generate an infinite hierarchy of conserved quantities, all in involution, for free. Given the two structures, define a sequence of functionals $H_0, H_1, H_2, \ldots$ recursively by

$$\mathcal{J}_1 \delta H_{n+1} = \mathcal{J}_0 \delta H_n.$$

(One has to show that this can always be solved for H_{n+1}; we take this on faith.) Then:

Theorem 4.11 (Magri)
Suppose $u_t = \mathcal{J}_0 \delta H_0 = \mathcal{J}_1 \delta H_1$ is bi-Hamiltonian, and let $\{H_n\}_{n \geq 0}$ be defined by the recursion above. Then all the H_n are first integrals, and they are in involution:

$$\{H_n, H_m\} = 0 \quad \text{for all } n, m \geq 0,$$

where the bracket is taken with respect to $\mathcal{J}_1$.

Proof. The whole proof rests on one observation. For $m \geq 1$,

$$\begin{aligned}
\{H_n, H_m\} &= \langle \delta H_n, \mathcal{J}_1 \delta H_m \rangle \\
&= \langle \delta H_n, \mathcal{J}_0 \delta H_{m-1} \rangle && \text{(recursion)} \\
&= -\langle \mathcal{J}_0 \delta H_n, \delta H_{m-1} \rangle && \text{($\mathcal{J}_0$ antisymmetric)} \\
&= -\langle \mathcal{J}_1 \delta H_{n+1}, \delta H_{m-1} \rangle && \text{(recursion)} \\
&= \langle \delta H_{n+1}, \mathcal{J}_1 \delta H_{m-1} \rangle && \text{($\mathcal{J}_1$ antisymmetric)} \\
&= \{H_{n+1}, H_{m-1}\}.
\end{aligned}$$

So we may shift an index from one slot to the other. Iterating m times gives

$$\{H_n, H_m\} = \{H_{n+m}, H_0\},$$

and iterating n times in the other direction gives $\{H_n, H_m\} = \{H_0, H_{n+m}\}$. Hence $\{H_n, H_m\} = \{H_m, H_n\}$; but the bracket is antisymmetric, so both sides vanish.

Finally, taking $m = 1$ shows $\{H_n, H_1\} = 0$ for all n, and H_1 is the Hamiltonian, so every H_n is a first integral. □

This is a genuinely miraculous piece of bookkeeping: the mere existence of a second Hamiltonian structure forces infinitely many commuting conservation laws into existence. For KdV the resulting H_n are (up to normalisation) the integrals $\int S_{2n+1}\, \mathrm{d}x$ we constructed from the scattering problem in Section 3.6. And the flows they generate,

$$u_{t_n} = \mathcal{J}_1 \delta H_n,$$

are the members of the KdV hierarchy: $n = 0$ gives $u_t = u_x$ (translation), $n = 1$ gives KdV, $n = 2$ gives a fifth-order equation, and so on. Since the Hamiltonians are in involution, all these flows commute, exactly as the Hamiltonian flows on a Liouville torus did in Section 1.

§4.3 Zero-Curvature Representation

There is a second, more geometric way to package integrability, which is often more flexible than the Lax formulation.

Given a function $u(x,t)$, suppose we can construct $N \times N$ matrices $U = U(\lambda)$ and $V = V(\lambda)$ depending on λ, on u and on its derivatives. Here λ is a **spectral parameter**, playing the same role as the λ in $L\varphi = \lambda\varphi$. Consider the linear system

$$\frac{\partial}{\partial x}\mathbf{v} = U(\lambda)\mathbf{v}, \qquad \frac{\partial}{\partial t}\mathbf{v} = V(\lambda)\mathbf{v}, \tag{$\dagger$}$$

for an N-component vector $\mathbf{v} = \mathbf{v}(x,t;\lambda)$.

This is an overdetermined system – we have twice as many equations as unknowns – so there is a compatibility condition, obtained from $\mathbf{v}_{xt} = \mathbf{v}_{tx}$:

$$\begin{aligned}
0 &= \frac{\partial}{\partial t}(U\mathbf{v}) - \frac{\partial}{\partial x}(V\mathbf{v}) \\
&= U_t\mathbf{v} + U\mathbf{v}_t - V_x\mathbf{v} - V\mathbf{v}_x \\
&= U_t\mathbf{v} + UV\mathbf{v} - V_x\mathbf{v} - VU\mathbf{v} \\
&= (U_t - V_x + [U,V])\,\mathbf{v}.
\end{aligned}$$

So if ($\dagger$) is to have a solution for every initial $\mathbf{v}_0$, we need

Definition 4.12 (Zero-curvature equation)
The **zero-curvature equation** is

$$\frac{\partial U}{\partial t} - \frac{\partial V}{\partial x} + [U,V] = 0.$$

A theorem of Frobenius says the converse also holds: if the zero-curvature equation is satisfied, then solutions of ($\dagger$) exist. So we have a correspondence between the solvability of a linear system and a nonlinear equation for U, V.

The name comes from differential geometry. The curvature of a connection A has components

$$F_{ij} = \frac{\partial A_j}{\partial x_i} - \frac{\partial A_i}{\partial x_j} + [A_i, A_j],$$

which is exactly the shape of our equation with $(A_1, A_2) = (U, V)$ and $(x_1, x_2) = (x, t)$. So the equation says that the connection built from U and V is flat.

Example 4.13 (Sine–Gordon)
Take

$$U(\lambda) = \frac{i}{2}\begin{pmatrix} 2\lambda & u_x \\ u_x & -2\lambda \end{pmatrix}, \qquad V(\lambda) = \frac{1}{4i\lambda}\begin{pmatrix} \cos u & -i\sin u \\ i\sin u & -\cos u \end{pmatrix}.$$

A computation shows that the zero-curvature equation is equivalent to

$$u_{xt} = \sin u,$$

the sine–Gordon equation in light-cone coordinates (Section 5.1).

The geometric picture pays off immediately. Curvature is an intrinsic object: if it vanishes in one gauge it vanishes in all of them. So given one pair (U, V) solving the zero-curvature equation, a gauge transformation

$$U \mapsto \tilde{U} = gUg^{-1} + g_x g^{-1}, \qquad V \mapsto \tilde{V} = gVg^{-1} + g_t g^{-1}$$

produces a new pair solving the same equation, and hence potentially a new integrable equation, or new solutions of the same one. This is explored on the last example sheet.

Zero curvature also supports its own version of inverse scattering. In the sine–Gordon example above, if $u_x \to 0$ as $|x| \to \infty$ then writing $\mathbf{v} = (\psi_1, \psi_2)^T$ the first equation of (†) becomes

$$\frac{\partial}{\partial x}\begin{pmatrix}\psi_1 \\ \psi_2\end{pmatrix} = i\lambda \begin{pmatrix}\psi_1 \\ -\psi_2\end{pmatrix},$$

so that

$$\begin{pmatrix}\psi_1 \\ \psi_2\end{pmatrix} \to A\begin{pmatrix}1 \\ 0\end{pmatrix}e^{i\lambda x} + B\begin{pmatrix}0 \\ 1\end{pmatrix}e^{-i\lambda x}$$

as $|x| \to \infty$. The coefficients A and B are the scattering data, and the second equation of (†) tells us how they evolve in t. This is the AKNS scheme, and it is how one solves sine–Gordon and the nonlinear Schrödinger equation.

§4.4 From Lax Pairs to Zero Curvature

The two formulations are closely related, and in fact every Lax pair gives rise to a zero-curvature representation.

Recall that in the isospectral flow theorem we found two equations holding simultaneously:

$$L\psi = \lambda\psi, \qquad \psi_t + A\psi = 0,$$

the second because $\tilde{\psi}$ vanished. Conversely, suppose we *impose* these two equations and demand $\lambda_t = 0$. Differentiating the first and substituting the second gives

$$L_t\psi + L\psi_t = \lambda\psi_t \implies L_t\psi = (L-\lambda)(-\psi_t) = (L-\lambda)A\psi = LA\psi - AL\psi,$$

that is $L_t = [L, A]$. So Lax's equation is exactly the compatibility condition for the pair of linear equations above – the same logical role played by the zero-curvature equation for (†).

To make the correspondence explicit, suppose

$$L = \partial_x^n + \sum_{j=0}^{n-1} u_j(x,t)\partial_x^j, \qquad A = \partial_x^m + \sum_{j=0}^{m-1} v_j(x,t)\partial_x^j.$$

The equation $L\psi = \lambda\psi$ lets us express every derivative of order $\geq n$ in terms of the lower ones:

$$\partial_x^n\psi = \lambda\psi - \sum_{j=0}^{n-1} u_j\partial_x^j\psi,$$

and differentiating this repeatedly handles the higher derivatives. So introducing the vector

$$\boldsymbol{\Psi} = \left(\psi, \partial_x\psi, \dots, \partial_x^{n-1}\psi\right)^T,$$

the equation $L\psi = \lambda\psi$ becomes a first-order system $\mathbf{\Psi}_x = U(\lambda)\mathbf{\Psi}$ with companion matrix

$$U(\lambda) = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ \lambda - u_0 & -u_1 & -u_2 & \cdots & -u_{n-1} \end{pmatrix}.$$

Similarly, differentiating $\psi_t + A\psi = 0$ a total of $i-1$ times with respect to x and using the above to eliminate high derivatives, we get

$$\left(\partial_x^{i-1}\psi\right)_t + \sum_{j=1}^{n} V_{ij}(x,t)\partial_x^{j-1}\psi = 0$$

for suitable V_{ij} built from the u_j, v_j and their derivatives, that is $\mathbf{\Psi}_t = -V\mathbf{\Psi}$ (after an overall sign). So

$$L_t = [L, A] \iff \begin{cases} L\psi = \lambda\psi \\ \psi_t + A\psi = 0 \end{cases} \iff \begin{cases} \mathbf{\Psi}_x = U(\lambda)\mathbf{\Psi} \\ \mathbf{\Psi}_t = V(\lambda)\mathbf{\Psi} \end{cases} \iff U_t - V_x + [U, V] = 0.$$

The zero-curvature formulation is strictly more general – there are integrable equations with a zero-curvature representation but no scalar Lax pair – and it is the one that generalises most readily to other settings.

§5 Further Integrable Equations

KdV is not alone. In this section we meet two more equations which are integrable in the same sense – they have Lax pairs, infinitely many conservation laws and soliton solutions – and which are at least as important in applications.

§5.1 The Sine–Gordon Equation

Definition 5.1 (Sine–Gordon equation)

The **sine–Gordon equation** is

$$u_{tt} - u_{xx} + \sin u = 0.$$

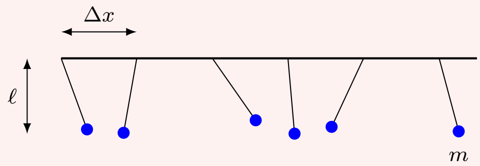
The name is a joke. The **Klein–Gordon equation** of relativistic quantum mechanics is

$$u_{tt} - u_{xx} + u = 0,$$

and replacing the u by $\sin u$ turns Klein into sine.

Example 5.2 (A chain of pendulums)

Here is a mechanical system governed by the equation. Take a horizontal rod, and hang from it a row of pendulums of length ℓ and mass m, spaced Δx apart, each free to rotate in the vertical plane perpendicular to the rod. Let $\theta_i(t)$ be the angle of the ith pendulum from the downward vertical, and connect neighbouring masses by torsional springs.



Gravity exerts a torque $-m\ell g \sin\theta_i$ on the ith pendulum, and the springs exert torques

$$\frac{K(\theta_{i+1}-\theta_i)}{\Delta x}, \qquad \frac{K(\theta_{i-1}-\theta_i)}{\Delta x}$$

from its two neighbours. Newton's second law for rotation, with moment of inertia $m\ell^2$, gives

$$m\ell^2 \frac{\mathrm{d}^2\theta_i}{\mathrm{d}t^2} = -mg\ell \sin\theta_i + \frac{K\left(\theta_{i+1} - 2\theta_i + \theta_{i-1}\right)}{\Delta x}.$$

Now divide by Δx and take the continuum limit $\Delta x \to 0$, holding $M = m/\Delta x$ fixed and writing $\theta_i(t) = \theta(i\Delta x, t)$. The second difference over Δx^2 tends to θ_{xx}, so

$$M\ell^2 \frac{\partial^2\theta}{\partial t^2} = -Mg\ell \sin\theta + K\frac{\partial^2\theta}{\partial x^2}.$$

Rescaling x and t to remove the constants gives $u_{tt} - u_{xx} + \sin u = 0$.

There is also a differential-geometric origin: solutions of the sine–Gordon equation correspond to *pseudospherical* surfaces in $\mathbb{R}^3$, that is surfaces of constant negative Gaussian curvature. This is the historical source of the equation, and of the Bäcklund transformations of Section 6.

It is often convenient to pass to **light-cone coordinates**

$$\xi = \tfrac{1}{2}(x-t), \qquad \tau = \tfrac{1}{2}(x+t),$$

in which $\partial_x = \frac{1}{2}(\partial_\xi + \partial_\tau)$ and $\partial_t = \frac{1}{2}(\partial_\tau - \partial_\xi)$, so that $\partial_t^2 - \partial_x^2 = -\partial_\xi\partial_\tau$. The equation becomes

$$\frac{\partial^2 u}{\partial\xi\partial\tau} = \sin u,$$

which is the form we will mostly use.

§5.1.1 Kinks and Antikinks

Let us find the travelling waves. Put $u(x,t) = f(\gamma(x-vt))$ with $|v| < 1$ and $\gamma = (1-v^2)^{-1/2}$, and write $\theta = \gamma(x-vt)$. Then $u_{tt} = v^2\gamma^2 f''$ and $u_{xx} = \gamma^2 f''$, so

$$\gamma^2(v^2-1)f'' + \sin f = 0, \qquad \text{i.e.} \qquad f'' = \sin f,$$

since $\gamma^2(v^2-1) = -1$. This is the equation of a pendulum with the sign of gravity reversed – which is no coincidence, given where the equation came from. Multiplying by f' and integrating,

$$\tfrac{1}{2}(f')^2 = -\cos f + C,$$

and imposing $f \to 0$, $f' \to 0$ as $\theta \to -\infty$ gives $C = 1$. Hence

$$(f')^2 = 2(1-\cos f) = 4\sin^2(f/2), \qquad f' = 2\sin(f/2),$$

taking the positive root. Separating variables and substituting $g = f/2$,

$$\int \frac{\mathrm{d}g}{\sin g} = \int \mathrm{d}\theta \quad \Longrightarrow \quad \log\tan\frac{g}{2} = \theta + \text{const},$$

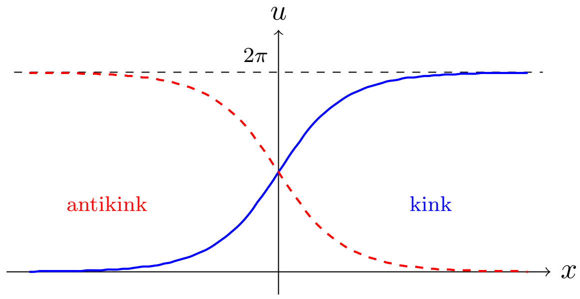
so $\tan(f/4) = Ae^{\theta}$. Absorbing A into a translation we obtain the following.

Proposition 5.3 (Kink solution)
For any $|v| < 1$,

$$u(x,t) = 4\tan^{-1}\left(\exp\left(\frac{x - vt}{\sqrt{1-v^2}}\right)\right)$$

solves the sine–Gordon equation. It is called a **kink**. Replacing $x - vt$ by $-(x - vt)$ gives the **antikink**.

The kink interpolates between $u = 0$ at $x = -\infty$ and $u = 2\pi$ at $x = +\infty$; the antikink runs the other way.



Note the Lorentz factor γ: a fast kink is narrower than a slow one, exactly as a moving rod is contracted in special relativity. This is not an accident, since the sine–Gordon equation is Lorentz invariant.

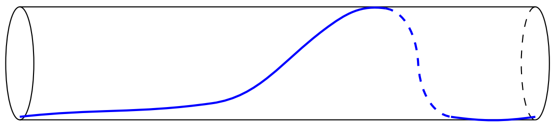
In the pendulum picture, a kink is a single full twist of the chain, propagating along it. Now remember that u is an *angle*: the values 0 and 2π describe the same physical configuration. So both the kink and the trivial solution $u = 0$ satisfy the same boundary conditions in the physical sense. Nevertheless the kink can never evolve into the trivial solution, and here is why.

Definition 5.4 (Topological charge)
For a solution with $u \to 2\pi n_{\pm}$ as $x \to \pm\infty$, the **topological charge** is

$$Q = \frac{1}{2\pi}\left[u\right]_{x=-\infty}^{x=+\infty} = n_+ - n_- \in \mathbb{Z}.$$

Since u evolves continuously and Q takes values in $\mathbb{Z}$, the charge cannot change: it is conserved for purely topological reasons, with no differential equation required. The kink has $Q = 1$, the antikink $Q = -1$, and the vacuum $Q = 0$. So a kink is stable, not because it sits at a minimum of some energy, but because there is no continuous path from it to anything else.

If we think of u as taking values on the circle S^1, the picture is that the solution is a path in the cylinder $\mathbb{R} \times S^1$ which winds Q times around:



No amount of continuous deformation, subject to the boundary conditions, can unwind it.

Remark. Sine–Gordon has multi-soliton solutions too, obtained (as we shall see in Section 6) by Bäcklund transformations. Besides kink–kink and kink–antikink scattering solutions there is a genuinely new object, the **breather**

$$u(x,t) = 4\tan^{-1}\left(\frac{\sqrt{1-\omega^2}}{\omega}\,\frac{\sin(\omega t)}{\cosh\left(\sqrt{1-\omega^2}\,x\right)}\right),$$

a kink and antikink bound together, oscillating in place with frequency ω. It has $Q = 0$, and yet it is perfectly stable – a periodic, localised, nonlinear excitation with no linear analogue.

§5.2 The Nonlinear Schrödinger Equation

Our final example is the equation governing envelope waves: pulses in optical fibres, wave packets on deep water, and Bose–Einstein condensates.

Definition 5.5 (Nonlinear Schrödinger equation)
The **nonlinear Schrödinger equation** (NLS) for a complex-valued $q(x,t)$ is

$$iq_t + q_{xx} + 2|q|^2 q = 0.$$

The sign of the nonlinear term matters. With $+2|q|^2q$ the equation is **focusing**, and supports *bright* solitons – localised pulses on a zero background. With $-2|q|^2q$ it is **defocusing**, and supports *dark* solitons – localised dips in a non-zero background. We treat the focusing case.

The equation looks like the linear Schrödinger equation of IB Quantum Mechanics with a potential $-2|q|^2$ generated by the wavefunction itself, which is exactly the physical situation in a nonlinear optical medium whose refractive index depends on the intensity of the light passing through it.

Let us find the soliton. Look first for a standing solution

$$q(x,t) = f(x)e^{i\theta t}$$

with f real. Then $iq_t = -\theta f e^{i\theta t}$, and the equation becomes

$$f'' = \theta f - 2f^3.$$

Multiplying by f' and integrating, with $f, f' \to 0$ at infinity,

$$(f')^2 = \theta f^2 - f^4 = f^2\left(\theta - f^2\right).$$

This has exactly the shape of the KdV travelling-wave equation, and the same guess works. Trying $f = a\,\mathrm{sech}(ax)$ we get

$$(f')^2 = a^4\,\mathrm{sech}^2(ax)\tanh^2(ax) = a^4\,\mathrm{sech}^2 - a^4\,\mathrm{sech}^4,$$

while

$$f^2(\theta - f^2) = a^2\theta \operatorname{sech}^2 - a^4 \operatorname{sech}^4,$$

so the two agree precisely when $\theta = a^2$.

Finally we set the soliton moving. NLS is Galilean invariant: if $q(x,t)$ is a solution then so is

$$\tilde{q}(x,t) = e^{i\left(\frac{v}{2}x - \frac{v^2}{4}t\right)} q(x - vt, t),$$

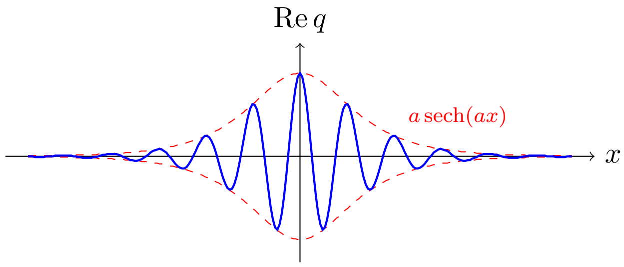
as one checks by direct substitution. Applying this to the standing solution gives the following.

Proposition 5.6 (Bright soliton)
For any $a > 0$ and $v \in \mathbb{R}$,

$$q(x,t) = a \operatorname{sech}\left(a(x - vt)\right) \exp\left(i\left[\frac{v}{2}x + \left(a^2 - \frac{v^2}{4}\right)t\right]\right)$$

solves the focusing NLS equation.

So the modulus $|q| = a \operatorname{sech}(a(x - vt))$ is a solitary wave of amplitude a and width $1/a$, travelling at speed v, while the phase oscillates underneath it:



Note the crucial difference from KdV: here the amplitude and the speed are *independent* parameters. A tall NLS soliton is narrow, but it need not be fast.

Like KdV, the NLS equation has a Lax pair and is solvable by inverse scattering; the relevant linear problem is the 2×2 Zakharov–Shabat system,

$$\frac{\partial}{\partial x}\begin{pmatrix}\psi_1 \\ \psi_2\end{pmatrix} = \begin{pmatrix}-i\lambda & q \\ -\bar{q} & i\lambda\end{pmatrix}\begin{pmatrix}\psi_1 \\ \psi_2\end{pmatrix},$$

which is exactly of the zero-curvature type $\mathbf{\Psi}_x = U(\lambda)\mathbf{\Psi}$ discussed in Section 4.3. Each discrete eigenvalue of this system corresponds to a soliton, with the real and imaginary parts of the eigenvalue giving the soliton's amplitude and velocity respectively. NLS solitons, like KdV solitons, pass through one another with nothing but a phase shift to show for it.

Finally, NLS is a Hamiltonian system in the sense of Section 4.1, with

$$iq_t = \frac{\delta H}{\delta \bar{q}}, \qquad H[q] = \int \left(|q_x|^2 - |q|^4\right)\,\mathrm{d}x,$$

and it has the conserved quantities

$$\int |q|^2\,\mathrm{d}x, \qquad \frac{i}{2}\int \left(q\bar{q}_x - \bar{q}q_x\right)\,\mathrm{d}x, \qquad H[q]$$

– the number of photons, the momentum, and the energy – together with infinitely many more.

§6 Bäcklund Transformations

For a linear PDE we have the principle of superposition: add two solutions and get a third. For a nonlinear PDE this fails completely, and it is precisely this failure that makes nonlinear equations hard. Bäcklund transformations are a partial substitute – a way of manufacturing new solutions from old, not by adding them, but by solving an auxiliary system of first-order equations.

The idea originated in differential geometry, where Bäcklund studied transformations taking one pseudospherical surface to another. We will only look at what it does to PDEs. The general definition is a little slippery, so we start with an example that will be familiar.

Example 6.1 (Cauchy–Riemann)

Consider the Cauchy–Riemann equations

$$u_x = v_y, \qquad u_y = -v_x.$$

From IB Complex Analysis we know that a pair (u, v) satisfies these if and only if both u and v are harmonic, that is

$$u_{xx} + u_{yy} = 0, \qquad v_{xx} + v_{yy} = 0.$$

(One direction is immediate: $u_{xx} + u_{yy} = (v_y)_x + (-v_x)_y = 0$.)

So suppose we have found some harmonic function v. We may then try to *solve* the Cauchy–Riemann equations for u, and whatever we get is guaranteed to be harmonic as well.

For instance take $v = 2xy$, which is harmonic. The Cauchy–Riemann equations become

$$u_x = 2x, \qquad u_y = -2y,$$

which we integrate to get

$$u(x, y) = x^2 - y^2 + C$$

for an arbitrary constant C. And indeed $x^2 - y^2$ is harmonic.

So the Cauchy–Riemann equations act as a machine for turning solutions of Laplace's equation into other solutions of Laplace's equation, at the cost of solving a first-order system. This is exactly what a Bäcklund transformation does.

Definition 6.2 (Bäcklund transformation)

A **Bäcklund transformation** is a system of equations relating solutions of one PDE to

(i) solutions of a *different* PDE; or

(ii) other solutions of the *same* PDE.

In the second case we call it an **auto-Bäcklund transformation**.

Transformations of the first kind are useful when one of the two equations is easier than the other.

Example 6.3 (The Liouville equation)
The equation $u_{xt} = e^u$ is related to the trivial equation $v_{xt} = 0$ by the Bäcklund transformation

$$u_x + v_x = \sqrt{2}\exp\left(\frac{u-v}{2}\right), \qquad u_t - v_t = \sqrt{2}\exp\left(\frac{u+v}{2}\right).$$

Verifying this is an exercise on the first example sheet. Since $v_{xt} = 0$ has the general solution $v = f(x) + g(t)$, this reduces the nonlinear Liouville equation to a pair of first-order ODEs.

§6.1 An Auto-Bäcklund Transformation for Sine–Gordon

The most important example, and the one we will use, is for the sine–Gordon equation in light-cone coordinates,

$$\varphi_{\xi\tau} = \sin\varphi.$$

Proposition 6.4
Fix a non-zero constant ε, and suppose φ and ψ are related by

$$\frac{\partial}{\partial\xi}(\varphi-\psi) = 2\varepsilon\sin\left(\frac{\varphi+\psi}{2}\right), \tag{B1}$$

$$\frac{\partial}{\partial\tau}(\varphi+\psi) = \frac{2}{\varepsilon}\sin\left(\frac{\varphi-\psi}{2}\right). \tag{B2}$$

Then φ solves the sine–Gordon equation if and only if ψ does.

Proof. Differentiate (B1) with respect to τ and use (B2):

$$\begin{aligned}(\varphi-\psi)_{\xi\tau} &= 2\varepsilon\cos\left(\frac{\varphi+\psi}{2}\right)\cdot\frac{1}{2}(\varphi+\psi)_\tau \\ &= \varepsilon\cos\left(\frac{\varphi+\psi}{2}\right)\cdot\frac{2}{\varepsilon}\sin\left(\frac{\varphi-\psi}{2}\right) \\ &= 2\cos\left(\frac{\varphi+\psi}{2}\right)\sin\left(\frac{\varphi-\psi}{2}\right) \\ &= \sin\varphi - \sin\psi,\end{aligned}$$

using the identity $2\cos A\sin B = \sin(A+B) - \sin(A-B)$.

Similarly, differentiating (B2) with respect to ξ and using (B1),

$$\begin{aligned}(\varphi+\psi)_{\xi\tau} &= \frac{2}{\varepsilon}\cos\left(\frac{\varphi-\psi}{2}\right)\cdot\frac{1}{2}(\varphi-\psi)_\xi \\ &= \frac{1}{\varepsilon}\cos\left(\frac{\varphi-\psi}{2}\right)\cdot 2\varepsilon\sin\left(\frac{\varphi+\psi}{2}\right) \\ &= \sin\varphi + \sin\psi.\end{aligned}$$

Adding the two displayed results gives $2\varphi_{\xi\tau} = 2\sin\varphi$, and subtracting them gives

> $2\psi_{\xi\tau} = 2\sin\psi$. So in fact both equations hold, and in particular each implies the other. □

Note that we have a whole *family* of auto-Bäcklund transformations, one for each $\varepsilon \neq 0$. This parameter will be essential.

§6.2 Generating Solitons from Nothing

Let us put the transformation to work. We start with the most boring solution imaginable, $\psi \equiv 0$, and see what comes out. The equations (B1), (B2) become

$$\varphi_\xi = 2\varepsilon \sin\frac{\varphi}{2}, \qquad \varphi_\tau = \frac{2}{\varepsilon}\sin\frac{\varphi}{2}.$$

There is an evident symmetry between ξ and τ here, which suggests the ansatz

$$\varphi(\xi,\tau) = 2\chi(z), \qquad z = \varepsilon\xi + \frac{\tau}{\varepsilon}.$$

Then $\varphi_\xi = 2\varepsilon\chi'$ and $\varphi_\tau = (2/\varepsilon)\chi'$, and *both* equations reduce to the same ODE,

$$\frac{\mathrm{d}\chi}{\mathrm{d}z} = \sin\chi.$$

This we can separate:

$$\int \frac{\mathrm{d}\chi}{\sin\chi} = \int \mathrm{d}z \quad \Longrightarrow \quad \log\tan\frac{\chi}{2} = z + \text{const},$$

so $\tan(\chi/2) = Ae^z$ and $\chi = 2\tan^{-1}(Ae^z)$. Therefore

$$\varphi(\xi,\tau) = 4\tan^{-1}\left(A\exp\left(\varepsilon\xi + \frac{\tau}{\varepsilon}\right)\right).$$

Let us see what this is. Substituting $\xi = \frac{1}{2}(x-t)$ and $\tau = \frac{1}{2}(x+t)$,

$$\varepsilon\xi + \frac{\tau}{\varepsilon} = \frac{\varepsilon^2+1}{2\varepsilon}x + \frac{1-\varepsilon^2}{2\varepsilon}t = \gamma\left(x - vt\right),$$

where

$$\gamma = \frac{\varepsilon^2+1}{2\varepsilon}, \qquad v = \frac{\varepsilon^2-1}{\varepsilon^2+1}.$$

A quick check confirms these are consistent with $\gamma = (1-v^2)^{-1/2}$:

$$1 - v^2 = \frac{(\varepsilon^2+1)^2 - (\varepsilon^2-1)^2}{(\varepsilon^2+1)^2} = \frac{4\varepsilon^2}{(\varepsilon^2+1)^2}.$$

So φ is exactly the kink of Section 5.1, and the Bäcklund parameter ε is nothing but a relabelling of the kink's velocity. We have generated a soliton out of the vacuum, by solving two first-order ODEs.

The obvious next move is to apply the transformation again, to the kink instead of the vacuum, and get a two-soliton solution. This works, but it requires solving another pair of PDEs, and the algebra gets unpleasant. Remarkably, there is a way to avoid it entirely.

§6.3 Bianchi Permutability

Suppose we start from a solution φ_0 and apply the Bäcklund transformation twice, with two different parameters ε_1 and ε_2, in the two possible orders. Write $\varphi_1 = B_{\varepsilon_1}(\varphi_0)$ and $\varphi_2 = B_{\varepsilon_2}(\varphi_0)$. The claim is that we may then choose $B_{\varepsilon_2}(\varphi_1)$ and $B_{\varepsilon_1}(\varphi_2)$ to be the *same* solution φ_{12} – the transformations commute:

$$\begin{tikzcd} & \varphi_{12} & \\ \varphi_1 \arrow[ur, "\varepsilon_2"] & & \varphi_2 \arrow[ul, "\varepsilon_1"'] \\ & \varphi_0 \arrow[ul, "\varepsilon_1"] \arrow[ur, "\varepsilon_2"'] & \end{tikzcd}$$

This is the **Bianchi permutability theorem**, and its real content is that φ_{12} can be written down *algebraically*, with no integration at all.

Theorem 6.5 (Bianchi permutability)
With notation as above,

$$\tan\left(\frac{\varphi_{12}-\varphi_0}{4}\right) = \frac{\varepsilon_1+\varepsilon_2}{\varepsilon_1-\varepsilon_2}\tan\left(\frac{\varphi_1-\varphi_2}{4}\right).$$

Proof. We use only (B1), applied along each of the four edges of the diagram:

$$(\varphi_1-\varphi_0)_\xi = 2\varepsilon_1\sin\left(\frac{\varphi_1+\varphi_0}{2}\right), \qquad (\varphi_2-\varphi_0)_\xi = 2\varepsilon_2\sin\left(\frac{\varphi_2+\varphi_0}{2}\right),$$
$$(\varphi_{12}-\varphi_1)_\xi = 2\varepsilon_2\sin\left(\frac{\varphi_{12}+\varphi_1}{2}\right), \qquad (\varphi_{12}-\varphi_2)_\xi = 2\varepsilon_1\sin\left(\frac{\varphi_{12}+\varphi_2}{2}\right).$$

Adding the first and third gives an expression for $(\varphi_{12}-\varphi_0)_\xi$; adding the second and fourth gives another. Equating them,

$$\varepsilon_1\sin\left(\frac{\varphi_1+\varphi_0}{2}\right)+\varepsilon_2\sin\left(\frac{\varphi_{12}+\varphi_1}{2}\right) = \varepsilon_2\sin\left(\frac{\varphi_2+\varphi_0}{2}\right)+\varepsilon_1\sin\left(\frac{\varphi_{12}+\varphi_2}{2}\right),$$

that is

$$\varepsilon_1\left[\sin\left(\frac{\varphi_1+\varphi_0}{2}\right)-\sin\left(\frac{\varphi_{12}+\varphi_2}{2}\right)\right] = \varepsilon_2\left[\sin\left(\frac{\varphi_2+\varphi_0}{2}\right)-\sin\left(\frac{\varphi_{12}+\varphi_1}{2}\right)\right].$$

Now apply $\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$ to each side. The crucial point is that the *same* quantity

$$\frac{A+B}{2} = \frac{\varphi_0+\varphi_1+\varphi_2+\varphi_{12}}{4}$$

appears on both sides, so the cosine factor cancels (generically it is non-zero). We are left with

$$\varepsilon_1 \sin\left(\frac{\varphi_0+\varphi_1-\varphi_2-\varphi_{12}}{4}\right) = \varepsilon_2 \sin\left(\frac{\varphi_0-\varphi_1+\varphi_2-\varphi_{12}}{4}\right).$$

Write

$$P = \frac{\varphi_0-\varphi_{12}}{4}, \qquad Q = \frac{\varphi_1-\varphi_2}{4},$$

so this reads $\varepsilon_1 \sin(P+Q) = \varepsilon_2 \sin(P-Q)$. Expanding both sides,

$$\varepsilon_1 \left(\sin P \cos Q + \cos P \sin Q\right) = \varepsilon_2 \left(\sin P \cos Q - \cos P \sin Q\right),$$

so

$$(\varepsilon_1-\varepsilon_2)\sin P \cos Q = -\left(\varepsilon_1+\varepsilon_2\right)\cos P \sin Q,$$

and dividing,

$$\tan P = -\frac{\varepsilon_1+\varepsilon_2}{\varepsilon_1-\varepsilon_2}\tan Q.$$

Since $P = -(\varphi_{12}-\varphi_0)/4$, this is the stated formula. □

This is a genuine **nonlinear superposition principle**. Given the vacuum $\varphi_0 = 0$ and two one-soliton solutions φ_1, φ_2, it delivers the two-soliton solution

$$\varphi_{12} = 4\tan^{-1}\left(\frac{\varepsilon_1+\varepsilon_2}{\varepsilon_1-\varepsilon_2}\tan\left(\frac{\varphi_1-\varphi_2}{4}\right)\right)$$

by pure algebra. Iterating produces the whole soliton hierarchy of sine–Gordon: from φ_{12} and φ_{13} we get φ_{123}, and so on. Every multi-soliton solution of the sine–Gordon equation can be reached this way, starting from nothing.

Example 6.6

Take $\varphi_0 = 0$ and two kinks

$$\varphi_j = 4\tan^{-1}\left(\exp\left(\gamma_j(x-v_jt)\right)\right), \qquad j = 1,2,$$

with parameters $\varepsilon_1, \varepsilon_2$ as above. Writing $E_j = \exp(\gamma_j(x-v_jt))$ and using

$$\tan\left(\frac{\varphi_1-\varphi_2}{4}\right) = \frac{\tan(\varphi_1/4)-\tan(\varphi_2/4)}{1+\tan(\varphi_1/4)\tan(\varphi_2/4)} = \frac{E_1-E_2}{1+E_1E_2},$$

the theorem gives the closed form

$$\varphi_{12}(x,t) = 4\tan^{-1}\left(\frac{\varepsilon_1+\varepsilon_2}{\varepsilon_1-\varepsilon_2}\cdot\frac{E_1-E_2}{1+E_1E_2}\right).$$

Depending on the signs and magnitudes of the parameters this describes kink–kink scattering, kink–antikink scattering, or – for complex conjugate ε's – the breather of Section 5.1. As with KdV, the only lasting effect of the collision is a phase shift.

Remark. It is no coincidence that we found a nonlinear superposition principle here and a determinant formula for KdV in Section 3.5. Both are shadows of the same underlying structure, and for KdV there is a Bäcklund transformation too, namely

$$w_{1,x} + w_{2,x} = \lambda - \tfrac{1}{2}\left(w_1-w_2\right)^2$$

relating potentials $u_j = w_{j,x}$. Its Bianchi permutability formula reproduces exactly the N-soliton solutions we obtained from the GLM equation.